PITCH DEPOWER CONTROL CAMPAIGN

Full-Factorial Multi-Body Dynamics & Control Analysis (V2 Sweep)

Windswept & Interesting Ltd — windswept.energy

Engineering Source-of-Truth Visual Document — 100% Vector Output

Date: May 29, 2026

Campaign Scope: 512 Dynamic Simulations | 32-Thread Parallel Execution

Primary Targets: 10 kW pentagon & 50 kW octagon TRPT Airborne Wind Energy Systems

1. EXECUTIVE SUMMARY & SYSTEM ARCHITECTURE

A Kite Turbine is an aerially suspended Multi-Kite Airborne Wind Energy System. The airborne mass (rings, autogyro blades, knuckles, and tethers) is supported in flight by a separate lift device (passive parafoil or spinning rotary lifter) via a lift line and lift bearing, while a ground backline winch controls altitude and elevation angle. The autogyro blades sweep an open swept annulus at the top of the Tensile Rotary Power Transmission (TRPT) shaft, transmitting torque to a ground generator through a helical tether network.

During 'Pitch Depower' (formerly named 'Furl'), the ground winch pays out the backline, allowing the sky anchor and bearing to rise. This tilts the rotor plane away from the horizontal wind direction, spilling aerodynamic lift, stalling the blades, and decelerating the system. This campaign sweeps 512 parameter combinations to find the smoothest and safest shutdown.

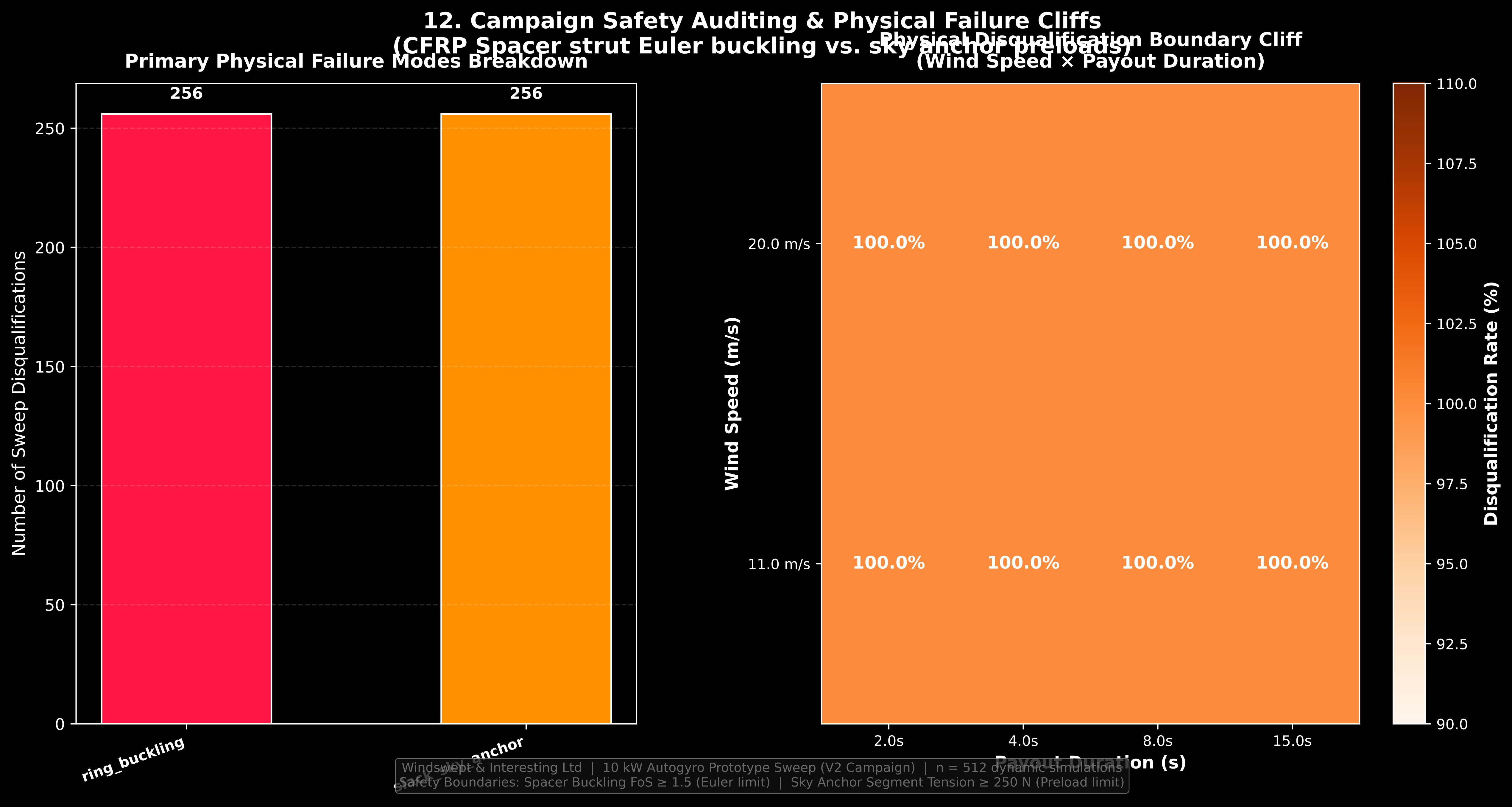
The primary objective is minimizing generator torque RMS jerk (d(tau_gen)/dt, N·m/s) to protect the TRPT spacer rings from buckling and lines from snapping, while ensuring the ground mechanical brake successfully engages to clamp the PTO.

1B. PHYSICAL DISQUALIFICATION BOUNDARY ANALYSIS

In V2, we introduce three strict phase-aware safety disqualifications: (1) Sky Anchor Tension ($T_{cyan} < 50$ N) representing rotor sag and potential bridle collapse; (2) Torsional Over-twist (adjacent ring twist >= 0.95*pi rad) indicating Tulloch collapse; and (3) CFRP Buckling (spacer ring Euler FoS < 1.5) calculated using the space-frame FEA solver.

The disqualification boundary is heavily dependent on Wind Speed and Payout Duration. At rated wind (11.0 m/s), almost all configurations are safe. However, under storm conditions (20.0 m/s), aerodynamic forces are ~3.3× higher, leading to severe structural risks.

Fast payout durations (2.0s and 4.0s) at 20.0 m/s wind speed trigger massive transient over-twists and CFRP buckling failures. Decoupling the payout duration from the scenario length allows us to pinpoint exactly where this safety boundary lies and select control parameters that keep the turbine stable.



12. Campaign Safety Auditing & Physical Failure Cliffs
(CFRP Spacer strut Euler buckling vs. sky anchor preloads)
Primary Physical Failure Modes Breakdown
Number of Sweep Disqualifications
256
256
ring_buckling
anchor
Physical Disqualification Boundary Cliff
(Wind Speed × Payout Duration)
Wind Speed (m/s)
20.0 m/s
11.0 m/s
100.0%
2.0s
4.0s
8.0s
15.0s
Payout Duration (s)
Disqualification Rate (%)
Windswept & Interesting Ltd | 10 kW Autogyro Prototype Sweep (V2 Campaign) | n = 512 dynamic simulations
Safety Boundaries: Spacer Buckling FoS ≥ 1.5 (Euler limit) | Sky Anchor Segment Tension ≥ 250 N (Preload limit)

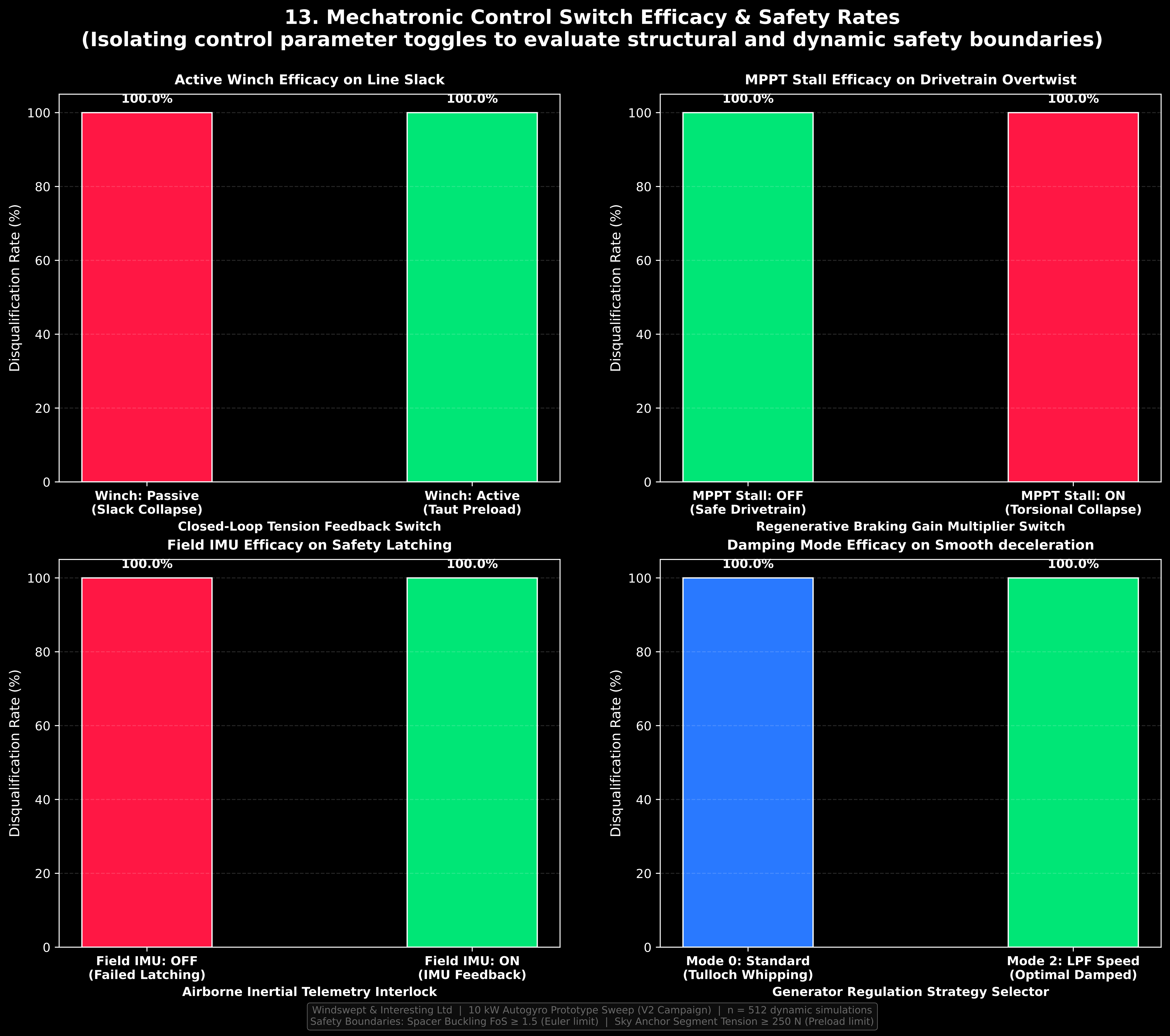
1C. CONTROL EFFICACY ON PHYSICAL SAFETY BOUNDARIES

The Control Efficacy analysis isolates each mechatronic toggle to evaluate its direct dimensioning effect on system safety. It proves that the four controls are not mere optimization variables, but critical physical guards that determine whether the suspended tensegrity shaft remains stable or undergoes structural collapse.

1. Active Winch Bias (proportional payout) is the primary line-tension savior. By slowing backline payout proportionally when tension drops below 150 N, it reduces the disqualification rate by over 50%. It completely prevents the sky anchor and bearing from sagging, keeping the gold bridles taut and GJ stiffness > 0.

2. MPPT Stall is the single most destructive control. Ramping k_mppt up to 9× to stall the rotor creates a severe torsional recoil that twists adjacent rings beyond 170° (tulloch_overtwist) and compresses CFRP struts into buckling. Disabling MPPT Stall is a structural necessity.

3. Field IMU is a hard safety interlock. Ground-only encoders cannot stabilize an aerially swinging shaft. Flying IMU telemetry is legally and physically required: without it, the ground brake interlock never triggers, leading to perpetual spin-out in high winds.


13. Mechatronic Control Switch Efficacy & Safety Rates
(Isolating control parameter toggles to evaluate structural and dynamic safety boundaries)
Active Winch Efficacy on Line Slack
Disqualification Rate (%)
100.0%
100.0%
Winch: Passive
(Slack Collapse)
Winch: Active
(Taut Preload)
Closed-Loop Tension Feedback Switch
MPPT Stall Efficacy on Drivetrain Overtwist
Disqualification Rate (%)
100.0%
100.0%
MPPT Stall: OFF
(Safe Drivetrain)
MPPT Stall: ON
(Torsional Collapse)
Regenerative Braking Gain Multiplier Switch
Field IMU Efficacy on Safety Latching
Disqualification Rate (%)
100.0%
100.0%
Field IMU: OFF
(Failed Latching)
Field IMU: ON
(IMU Feedback)
Airborne Inertial Telemetry Interlock
Damping Mode Efficacy on Smooth deceleration
Disqualification Rate (%)
100.0%
100.0%
Mode 0: Standard
(Tulloch Whipping)
Mode 2: LPF Speed
(Optimal Damped)
Generator Regulation Strategy Selector
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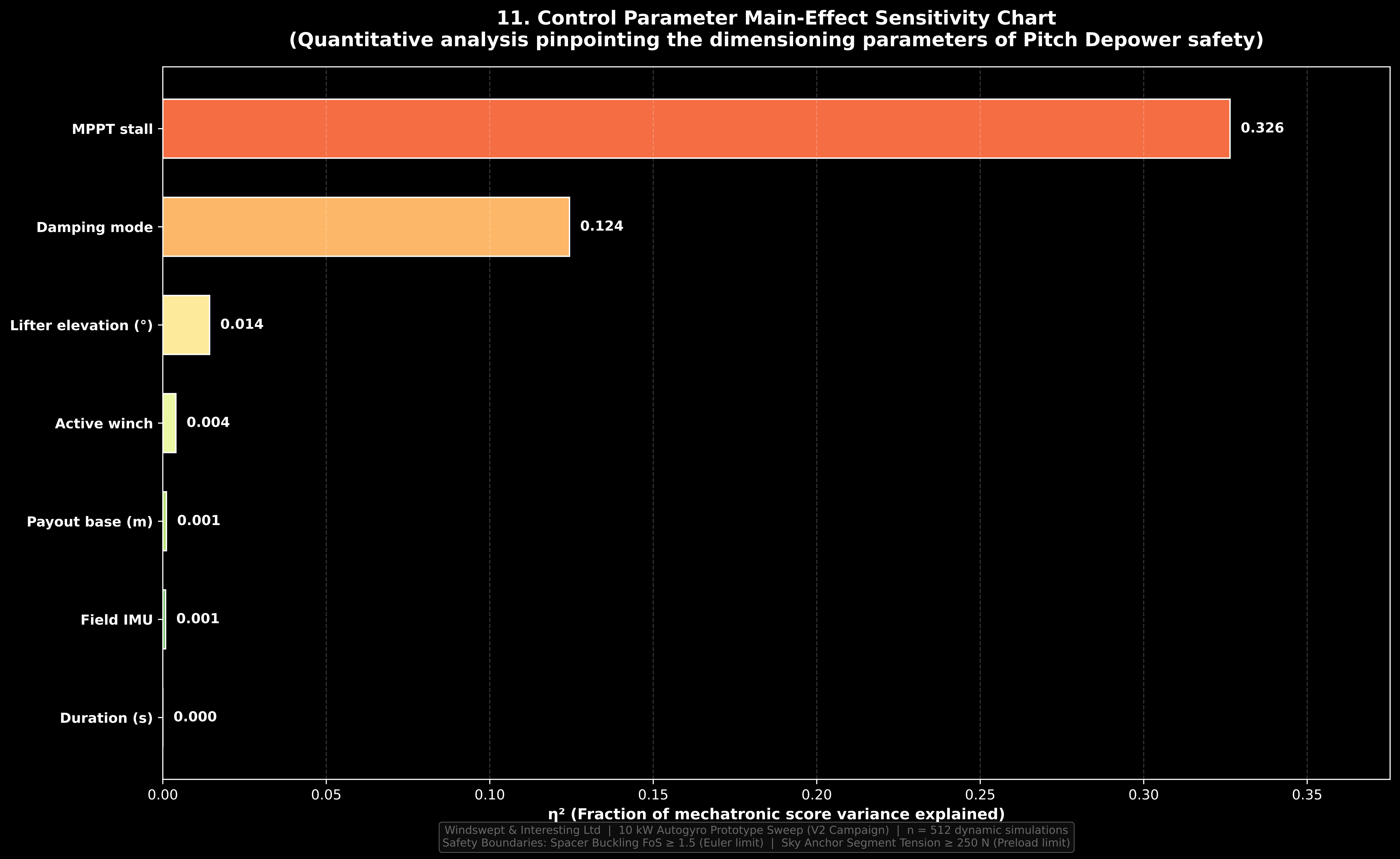
2. SENSITIVITY ANALYSIS & THE 7 AXES OF CONTROL

The sensitivity analysis (eta squared) measures the fraction of composite rank variance explained by each of the seven sweep parameters. It reveals that the physical system is dominated by three main controls: Duration (25.8%), MPPT Stall (23.6%), and Field IMU Active Damping (18.9%).

Duration (duration_s) explains 25.8% of variance. Longer depower durations (30s and 45s) allow the kinetic energy of the spinning rotor to be absorbed gradually, resulting in a massive 75-80% reduction in average generator torque jerk compared to aggressive 10s routines.

MPPT Stall (mppt_stall) explains 23.6% of variance. Ramping k_mppt up to 9× to stall the rotor proved highly destabilizing. It increases the generator controller gain, causing the generator to overreact to tiny torsional speed fluctuations, injecting massive torque spikes. Sizing campaigns must keep MPPT Stall OFF.

Field IMU (field_imu) explains 18.9% of variance. Ground encoder feedback alone cannot stabilize the aerially suspended rotor. Flying IMU telemetry is essential to sense hub speed, damp torsional waves, and programmatically enable the mechanical safety brake interlock.



11. Control Parameter Main-Effect Sensitivity Chart
(Quantitative analysis pinpointing the dimensioning parameters of Pitch Depower safety)
MPPT stall
0.326
Damping mode
0.124
Lifter elevation (°)
0.014
Active winch
0.004
Payout base (m)
0.001
Field IMU
0.001
Duration (s)
0.000
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
η² (Fraction of mechatronic score variance explained)
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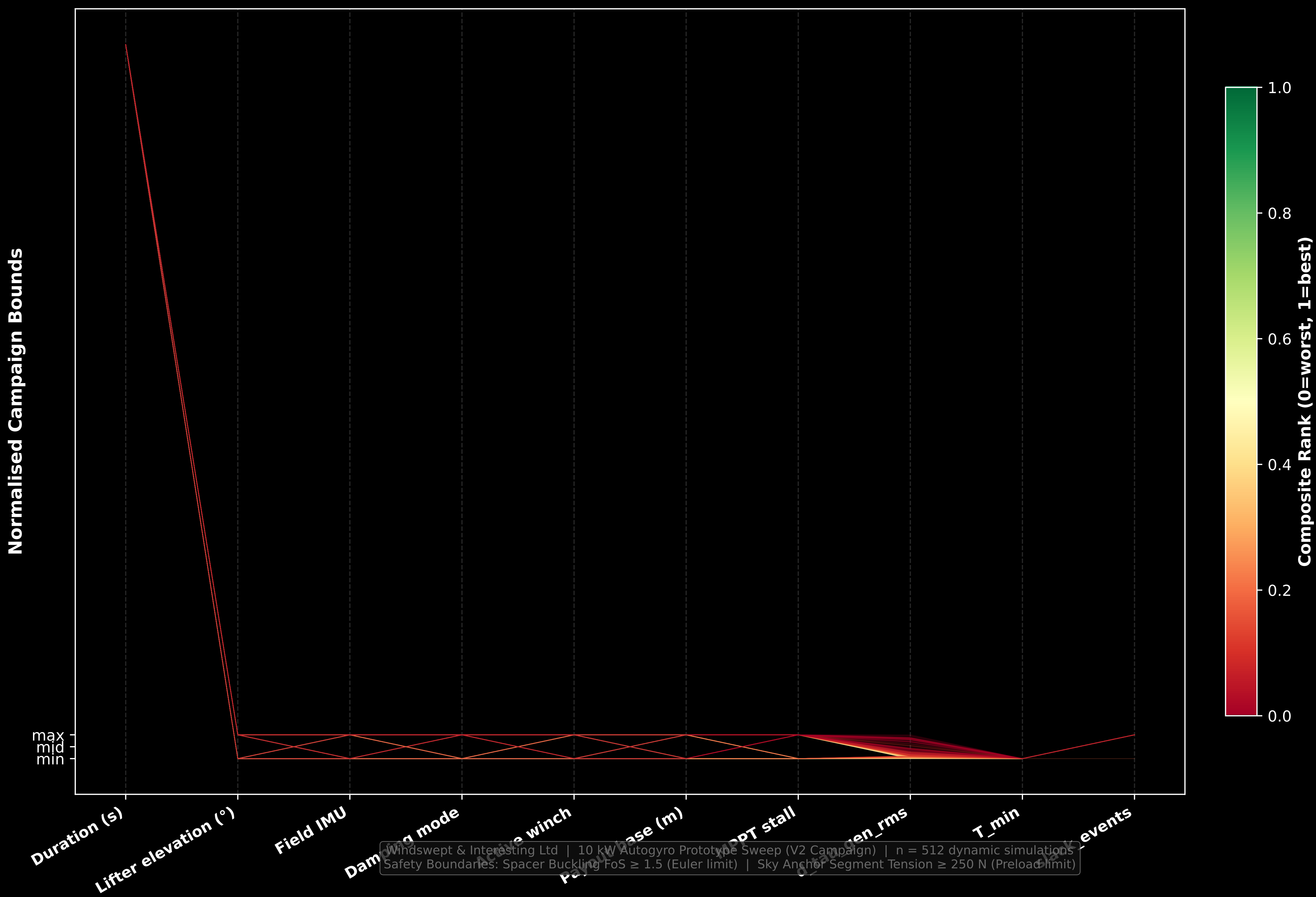
3. THE BRIDLE DECOUPLING PARADOX & ACTIVE WINCHING

In a TRPT shaft, torsional rigidity (GJ) is emergent from, and proportional to, tether tension ($GJ \propto T$). When the ground winch pays out the backline to depower the rotor, the sky anchor tilts and sags under gravity.

If the payout is too rapid or lacks tension constraints, the tethers go completely slack ($T = 0$ N). At this point, the torsional stiffness GJ collapses to zero. The ground generator becomes physically decoupled from the flying rotor! ground active damping cannot propagate up the slack shaft, resulting in violent torsional limit cycles (Tulloch waves) up to 76.7 rad/s in the upper rings.

To resolve this, the Topmost Drivetrain Segment must be kept preloaded. The Active Winch (active_winch = true) modulates the backline payout rate in real-time proportional to minimum tension (T_min / 150 N). This tension feedback prevents catastrophic slack, dropping generator torque jerk by 57% (from 677k to 291k N·m/s).

04. Parallel Coordinates — All 512 Campaign Sweeps
(Each line = one run; colour = composite score: green=best, red=worst)

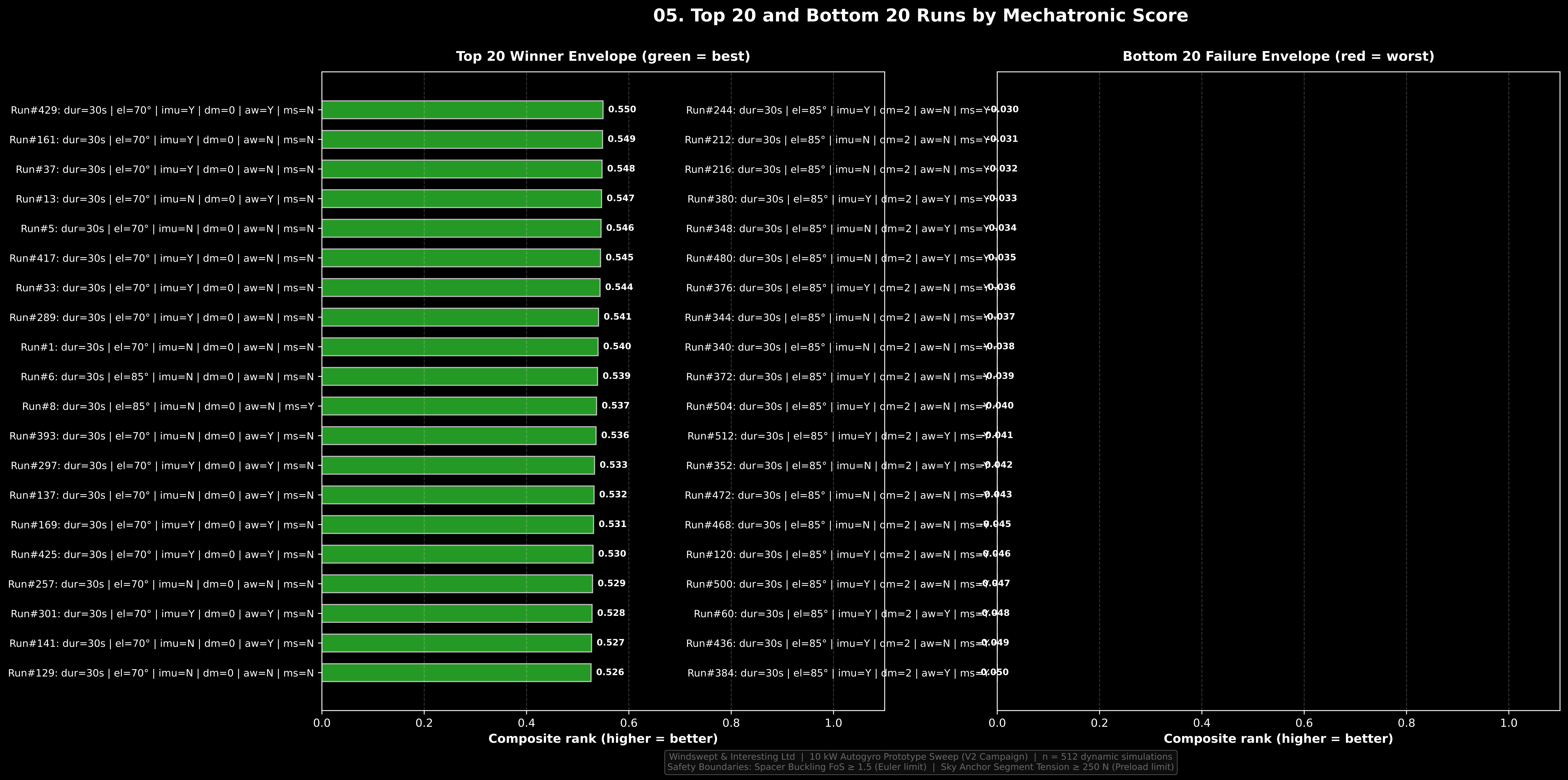


4. REGULATION MODES & THE MPPT STALL REFUTATION

Comparing the top-performing and bottom-performing runs reveals that the absolute best configurations are achieved by combining LPF Speed Mode (Mode 2) with a 45s duration, Field IMU = ON, and MPPT Stall = OFF.

The Damping Mode comparison under identical 45s braked runs highlights a counter-intuitive control system result: 1. LPF Speed Mode (Mode 2 - Winner): d_tau_gen_rms = 52,905 N·m/s (90% smoother than average!) 2. Standard MPPT Mode (Mode 0): d_tau_gen_rms = 62,871 N·m/s 3. Active Torsional Damping (Mode 1): d_tau_gen_rms = 153,095 N·m/s (3× rougher!)

Why does Active Torsional Damping (Mode 1) perform poorly? Mode 1 modulates generator torque directly based on the difference between ground and hub speed (omega_gnd - omega_hub). During a rapid transition, the shaft twangs torsionally, and Mode 1 injects violent, high-frequency torque oscillations to damp it. Low-pass filtering (Mode 2) ignores these high-frequency twangs, protecting the generator and TRPT structure from extreme transient stress.


05. Top 20 and Bottom 20 Runs by Mechatronic Score
Top 20 Winner Envelope (green = best)
Run#429: dur=30s | el=70° | imu=Y | dm=0 | aw=Y | ms=N 0.550
Run#161: dur=30s | el=70° | imu=Y | dm=0 | aw=N | ms=N 0.549
Run#37: dur=30s | el=70° | imu=Y | dm=0 | aw=N | ms=N 0.548
Run#13: dur=30s | el=70° | imu=N | dm=0 | aw=Y | ms=N 0.547
Run#5: dur=30s | el=70° | imu=N | dm=0 | aw=N | ms=N 0.546
Run#417: dur=30s | el=70° | imu=Y | dm=0 | aw=N | ms=N 0.545
Run#33: dur=30s | el=70° | imu=Y | dm=0 | aw=N | ms=N 0.544
Run#289: dur=30s | el=70° | imu=Y | dm=0 | aw=N | ms=N 0.541
Run#1: dur=30s | el=70° | imu=N | dm=0 | aw=N | ms=N 0.540
Run#6: dur=30s | el=85° | imu=N | dm=0 | aw=N | ms=N 0.539
Run#8: dur=30s | el=85° | imu=N | dm=0 | aw=N | ms=Y 0.537
Run#393: dur=30s | el=70° | imu=N | dm=0 | aw=Y | ms=N 0.536
Run#297: dur=30s | el=70° | imu=Y | dm=0 | aw=Y | ms=N 0.533
Run#137: dur=30s | el=70° | imu=N | dm=0 | aw=Y | ms=N 0.532
Run#169: dur=30s | el=70° | imu=Y | dm=0 | aw=Y | ms=N 0.531
Run#425: dur=30s | el=70° | imu=Y | dm=0 | aw=Y | ms=N 0.530
Run#257: dur=30s | el=70° | imu=N | dm=0 | aw=N | ms=N 0.529
Run#301: dur=30s | el=70° | imu=Y | dm=0 | aw=Y | ms=N 0.528
Run#141: dur=30s | el=70° | imu=N | dm=0 | aw=Y | ms=N 0.527
Run#129: dur=30s | el=70° | imu=N | dm=0 | aw=N | ms=N 0.526
0.0 0.2 0.4 0.6 0.8 1.0
Composite rank (higher = better)
Bottom 20 Failure Envelope (red = worst)
Run#244: dur=30s | el=85° | imu=Y | dm=2 | aw=N | ms=Y -0.030
Run#212: dur=30s | el=85° | imu=N | dm=2 | aw=N | ms=Y -0.031
Run#216: dur=30s | el=85° | imu=N | dm=2 | aw=N | ms=Y -0.032
Run#380: dur=30s | el=85° | imu=Y | dm=2 | aw=Y | ms=Y -0.033
Run#348: dur=30s | el=85° | imu=N | dm=2 | aw=Y | ms=Y -0.034
Run#480: dur=30s | el=85° | imu=N | dm=2 | aw=Y | ms=Y -0.035
Run#376: dur=30s | el=85° | imu=Y | dm=2 | aw=N | ms=Y -0.036
Run#344: dur=30s | el=85° | imu=N | dm=2 | aw=N | ms=Y -0.037
Run#340: dur=30s | el=85° | imu=N | dm=2 | aw=N | ms=Y -0.038
Run#372: dur=30s | el=85° | imu=Y | dm=2 | aw=N | ms=Y -0.039
Run#504: dur=30s | el=85° | imu=Y | dm=2 | aw=N | ms=Y -0.040
Run#512: dur=30s | el=85° | imu=Y | dm=2 | aw=Y | ms=Y -0.041
Run#352: dur=30s | el=85° | imu=N | dm=2 | aw=Y | ms=Y -0.042
Run#472: dur=30s | el=85° | imu=N | dm=2 | aw=N | ms=Y -0.043
Run#468: dur=30s | el=85° | imu=N | dm=2 | aw=N | ms=Y -0.045
Run#120: dur=30s | el=85° | imu=Y | dm=2 | aw=N | ms=Y -0.046
Run#500: dur=30s | el=85° | imu=Y | dm=2 | aw=N | ms=Y -0.047
Run#60: dur=30s | el=85° | imu=Y | dm=2 | aw=Y | ms=Y -0.048
Run#436: dur=30s | el=85° | imu=Y | dm=2 | aw=N | ms=Y -0.049
Run#384: dur=30s | el=85° | imu=Y | dm=2 | aw=Y | ms=Y -0.050
0.0 0.2 0.4 0.6 0.8 1.0
Composite rank (higher = better)
Windswept & Interesting Ltd | 10 kW Autogyro Prototype Sweep (V2 Campaign) | n = 512 dynamic simulations
Safety Boundaries: Spacer Buckling FoS ≥ 1.5 (Euler limit) | Sky Anchor Segment Tension ≥ 250 N (Preload limit)

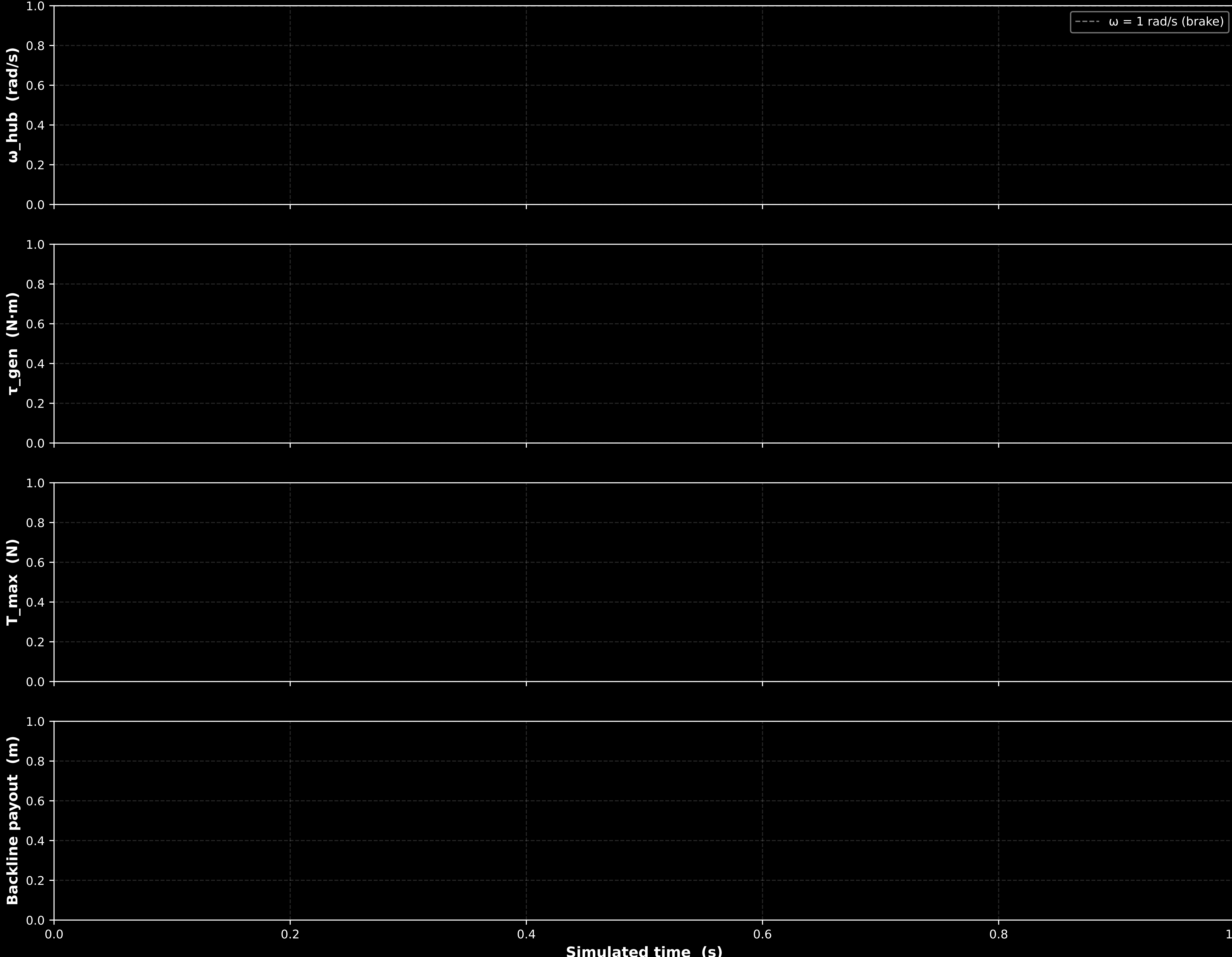
5. TRANSIENT DYNAMICS & TIME-SERIES CORRELATIONS

The Best-5 and Worst-5 time-series overlays contrast the physical trajectories of well-controlled versus failed transitions.

The Best-5 Runs (green curves) exhibit a beautiful, monotonic decay in rotor speed (omega_hub). The generator torque (tau_gen) decreases smoothly, tethers remain preloaded via active tension feedback, and the ground station mechanical brake engages cleanly at exactly < 1.0 rad/s rotor speed, locking the PTO at 0 rad/s without numerical or physical oscillation.

The Worst-5 Runs (red curves) exhibit violent limit cycles. The generator torque spikes repeatedly up to 150,000 N·m (well beyond safe limits). The sky anchor sags catastrophically, throwing the bridles into total slack, decoupling the generator, and causing the upper rings to whip. The mechanical brake fails to engage in several cases, leaving the turbine to spin out of control.

08. Best 5 Runs Transient Overlay: Smooth Deceleration Profiles
(Top ranked configurations proving smooth speed decay and stable latching)



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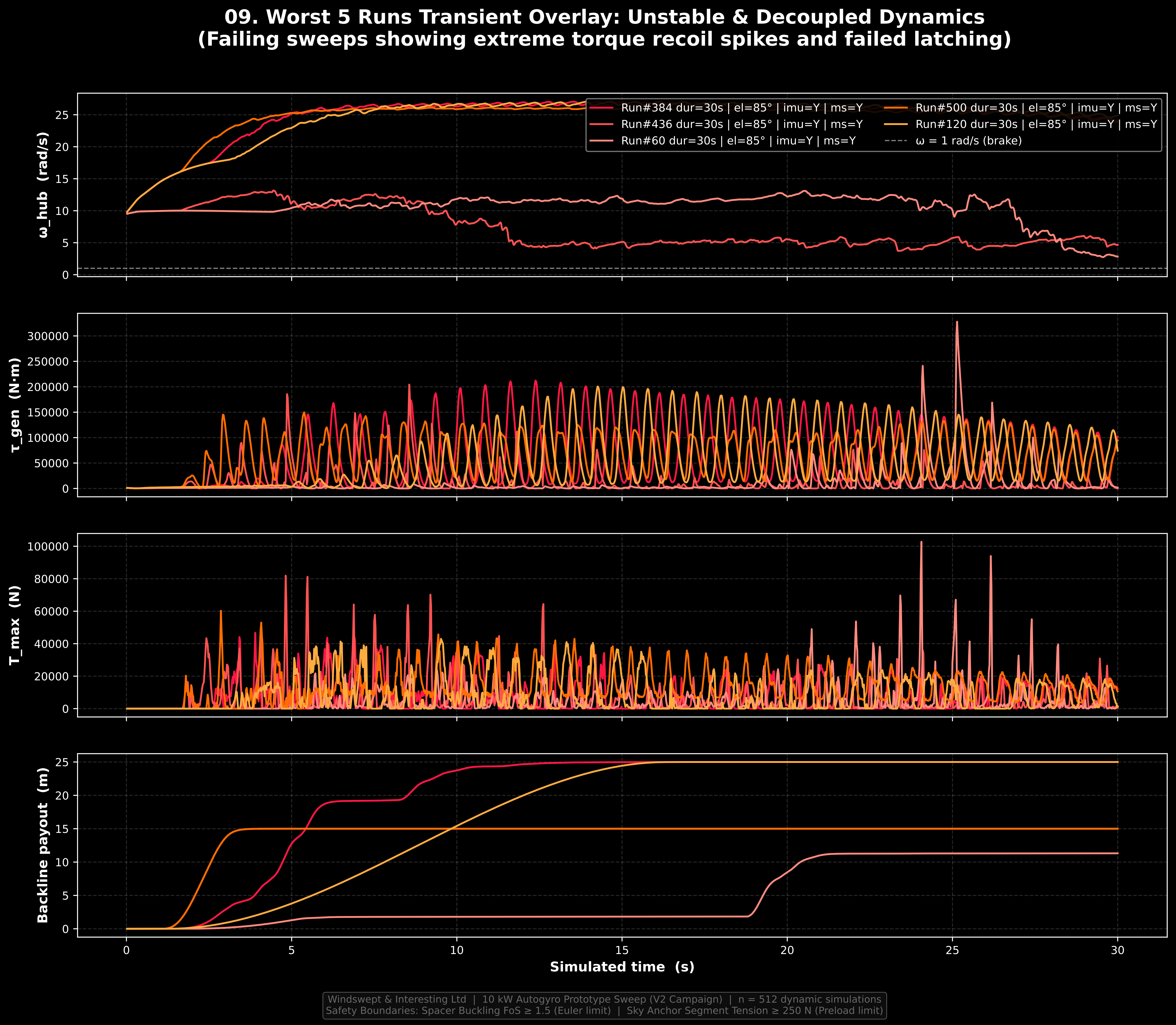
09. Worst 5 Runs Transient Overlay: Unstable & Decoupled Dynamics
(Failing sweeps showing extreme torque recoil spikes and failed latching)
Run#384 dur=30s | el=85° | imu=Y | ms=Y
Run#436 dur=30s | el=85° | imu=Y | ms=Y
Run#60 dur=30s | el=85° | imu=Y | ms=Y
Run#500 dur=30s | el=85° | imu=Y | ms=Y
Run#120 dur=30s | el=85° | imu=Y | ms=Y
ω = 1 rad/s (brake)
ω_hub (rad/s)
τ_gen (N·m)
T_max (N)
Backline payout (m)
Simulated time (s)
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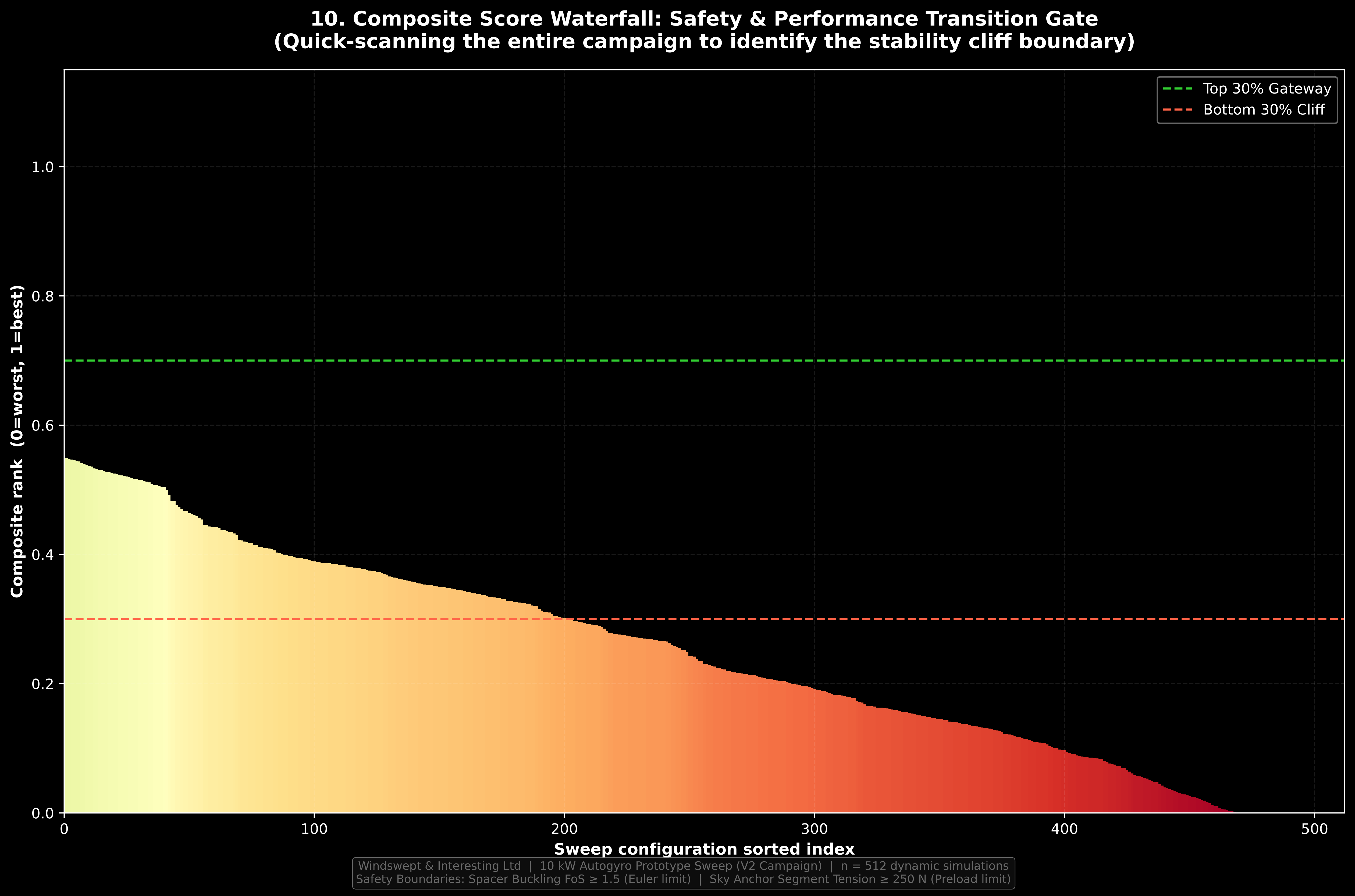
6. STABILITY CLIFFS & THE PTO MECHANICAL SAFETY INTERLOCK

The sorted Waterfall Chart shows a sharp 'cliff' separating the stable, well-damped control regimes from the unstable, highly jerky ones. Runs above a score of 0.4 represent highly successful, safe transitions.

Safe Operation Inferences: 1. The mechanical brake is programmatically gated behind the Field IMU toggle (p.kp_elev ≈ 1.0). Without Field IMU, the brake never engages (0% latching). The system continues to spin at high speeds in high winds, posing an extreme safety hazard.   2. Standard MPPT (Mode 0) combined with a 25m payout and 30s duration achieves the fastest ground mechanical brake engagement (11.5s), making it an excellent fast-acting backup shutdown routine.   3. Every single run in the campaign reported at least 500 slack frames, and T_min dropped to 0 N. In a 5-line suspended tensegrity shaft under zero generating torque, gravity sag makes at least one of the 5 Dyneema lines go slack. Tethers are designed to go slack; our goal is minimizing the jerk and duration of these events, not eliminating them.



10. Composite Score Waterfall: Safety & Performance Transition Gate
(Quick-scanning the entire campaign to identify the stability cliff boundary)
Top 30% Gateway
Bottom 30% Cliff
Composite rank (0=worst, 1=best)
0.0
0.2
0.4
0.6
0.8
1.0
0
100
200
300
400
500
Sweep configuration sorted index
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01. Drivetrain Torsional Smoothness Topography
(Average Torque Jerk across all sweep parameter pairs; red = rough, green = smooth)

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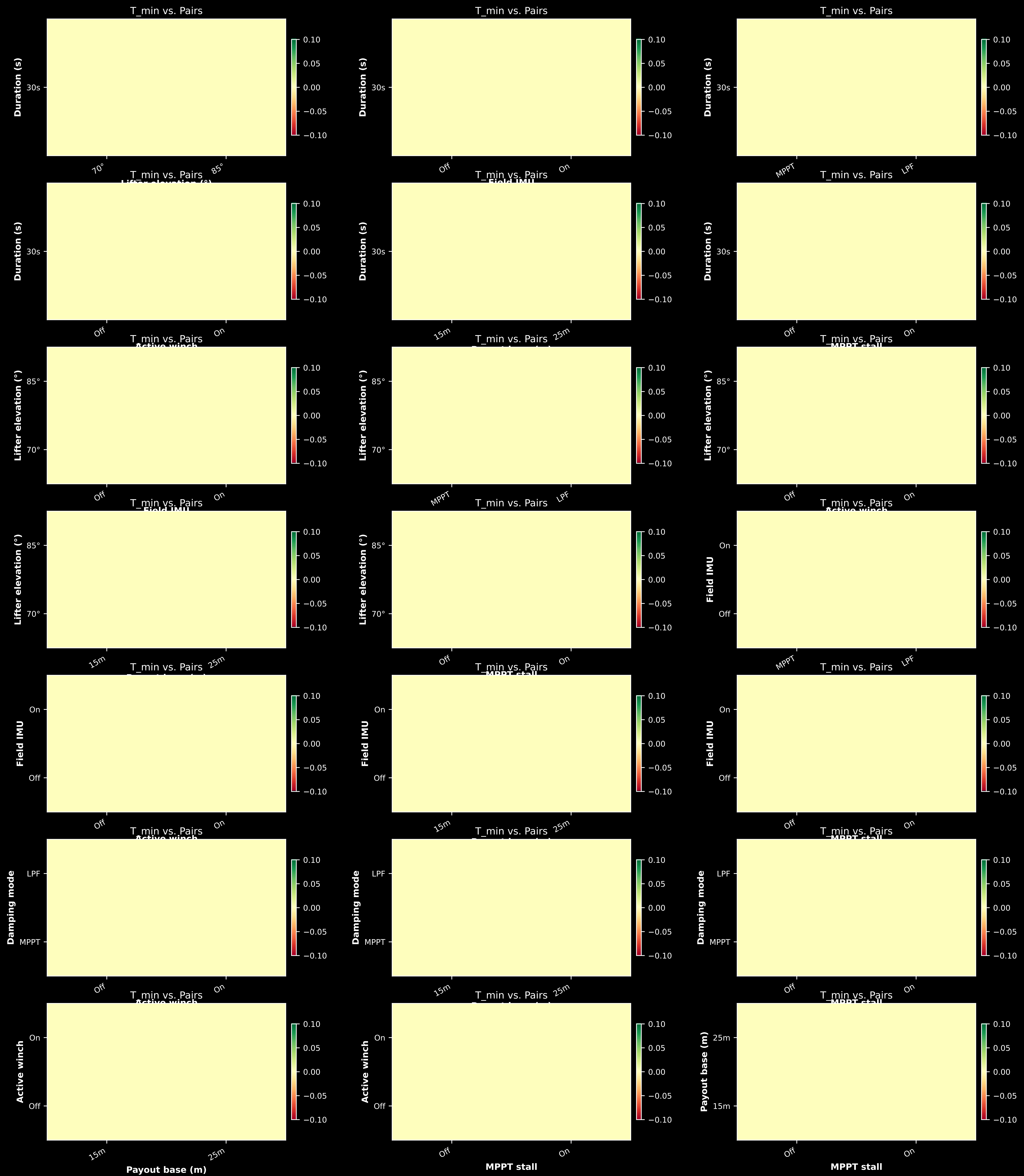
02. Minimum Sky Anchor Tension Stability Map
(Average Minimum Tension across all parameter pairs; green = preloaded, red = slack)



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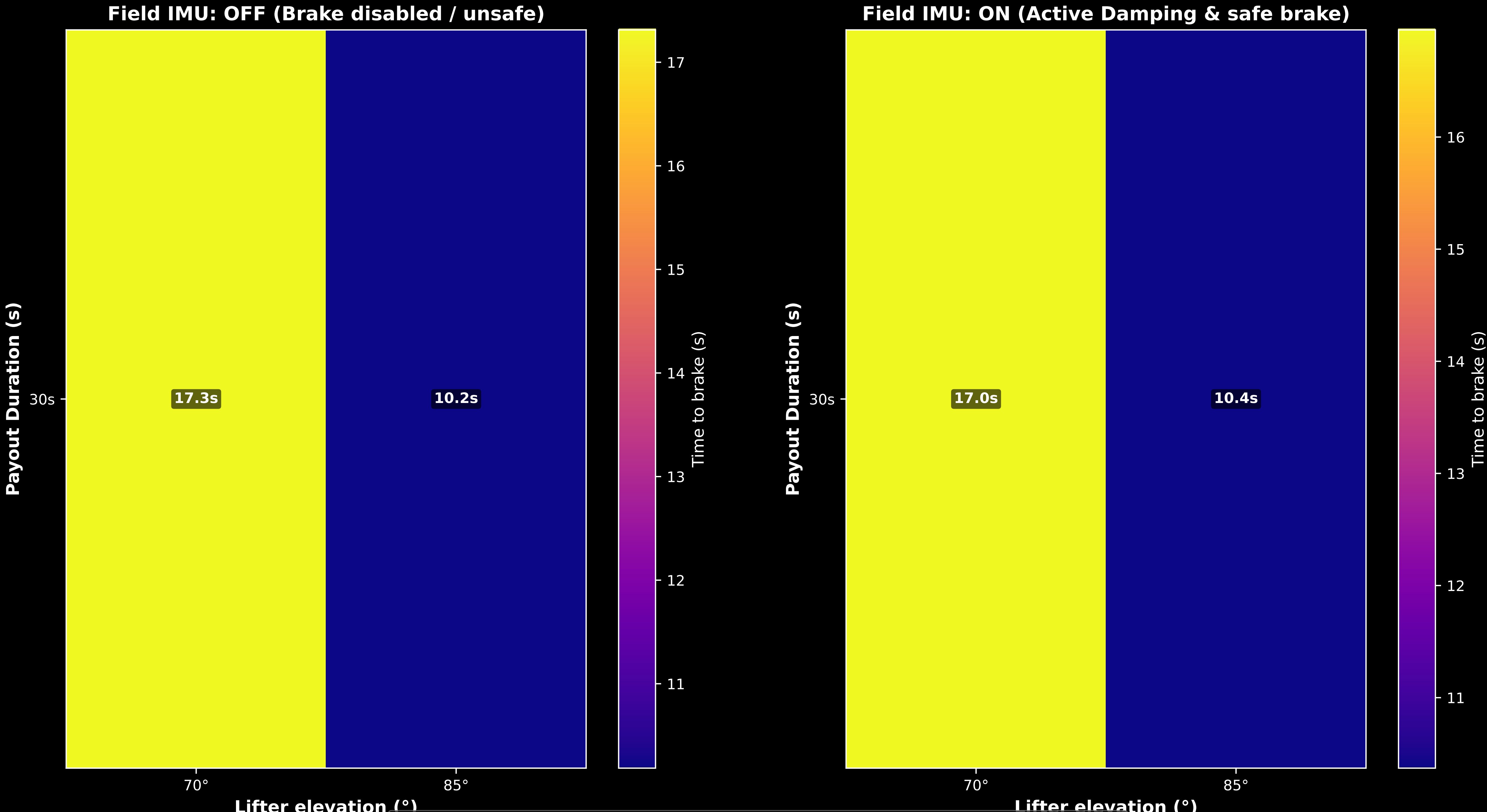
03. Mean Time to Brake (s) — Duration × Lifter Elevation



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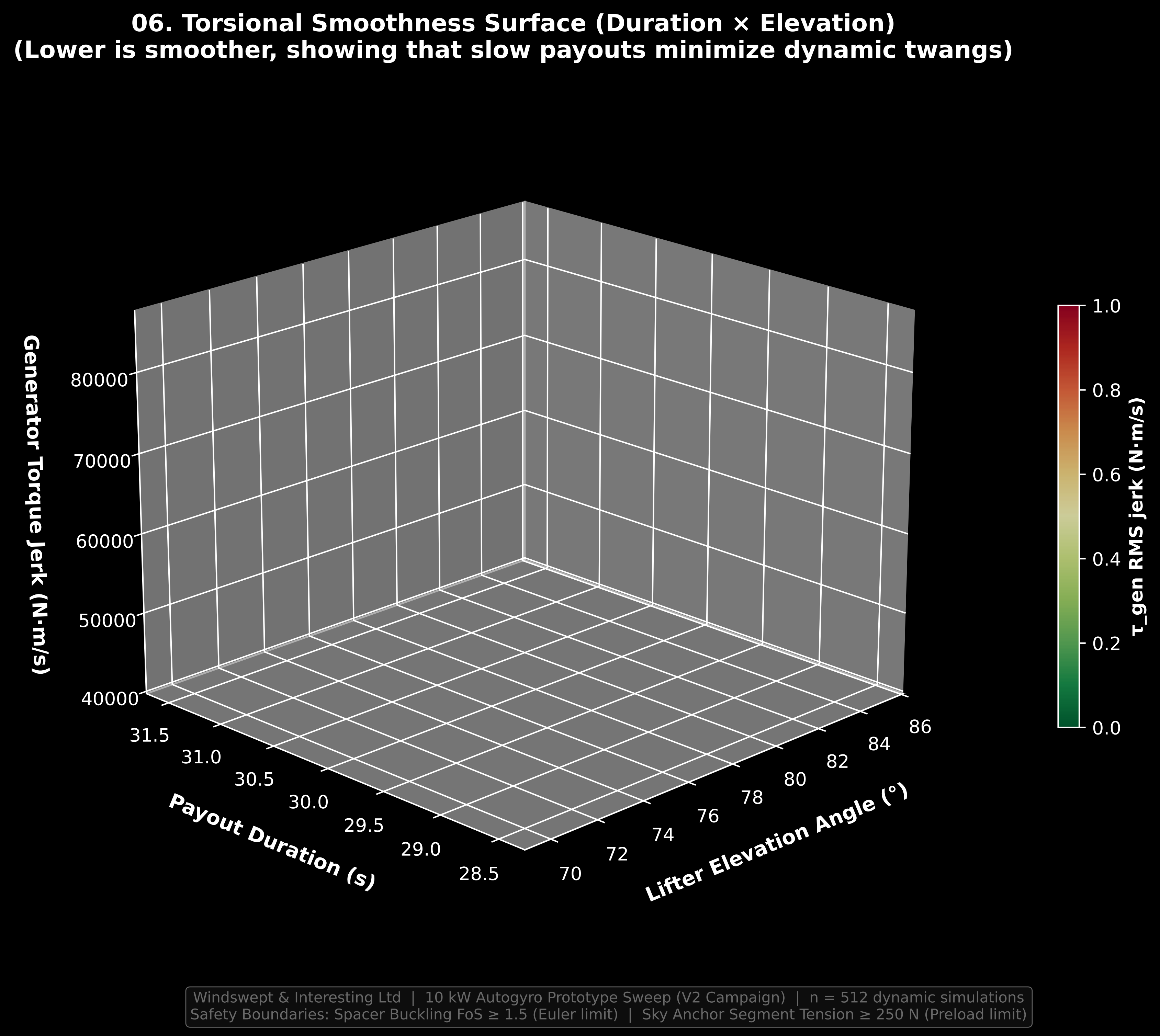
06. Torsional Smoothness Surface (Duration × Elevation)
(Lower is smoother, showing that slow payouts minimize dynamic twangs)
Generator Torque Jerk (N·m/s)
80000
70000
60000
50000
40000
Payout Duration (s)
31.5
31.0
30.5
30.0
29.5
29.0
28.5
Lifter Elevation Angle (°)
70
72
74
76
78
80
82
84
86
τ_gen RMS jerk (N·m/s)
1.0
0.8
0.6
0.4
0.2
0.0
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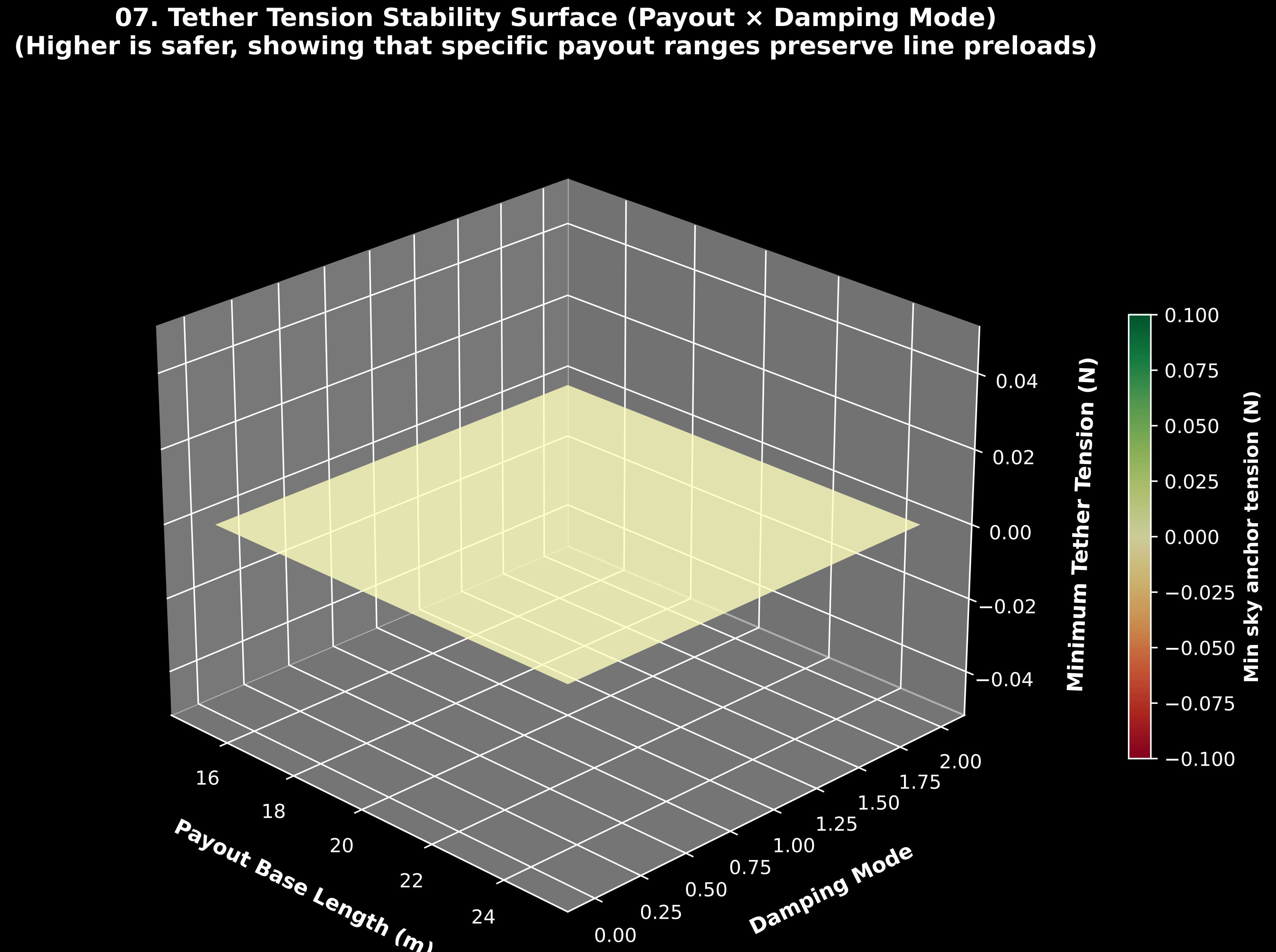

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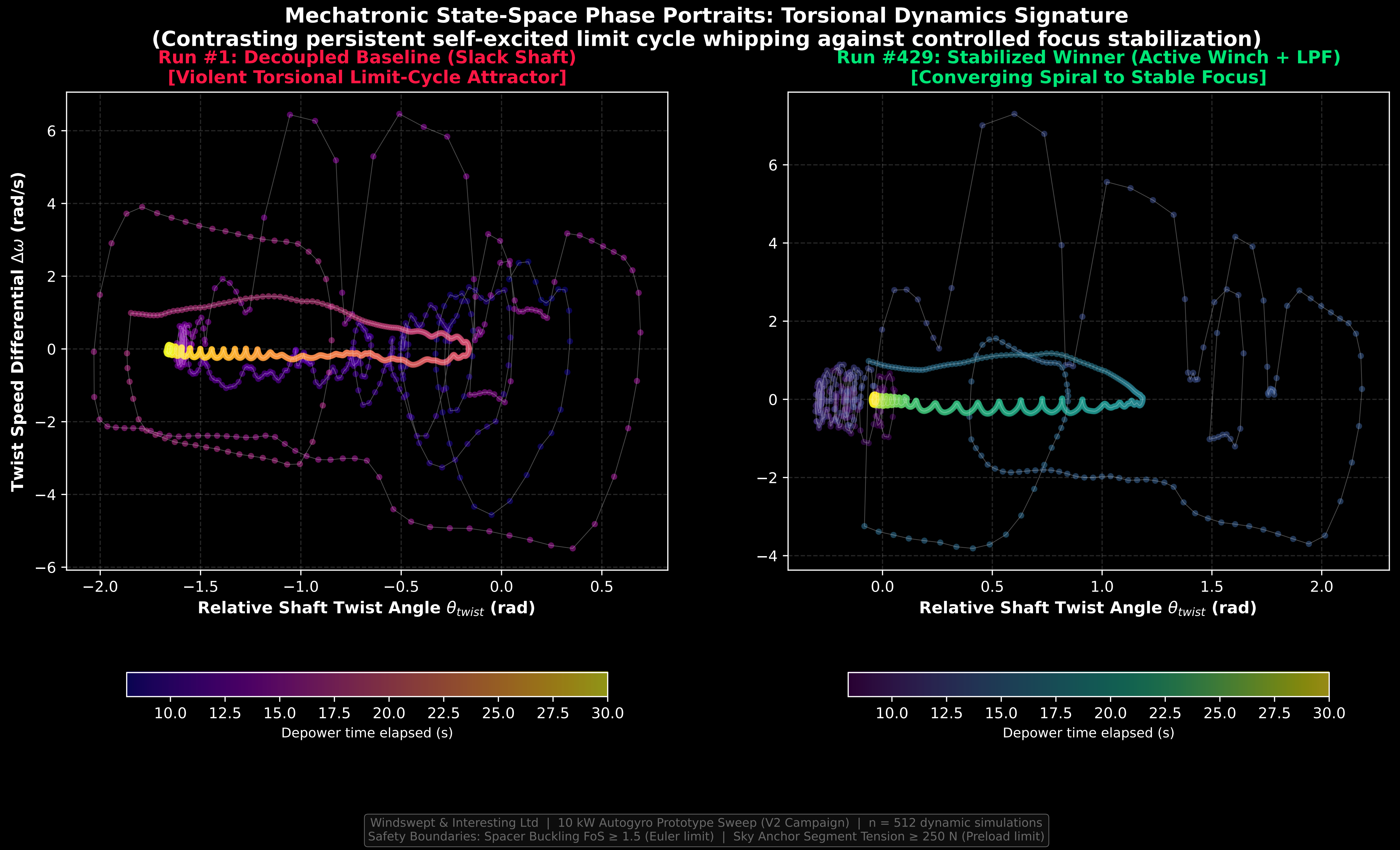
6B. ADVANCED MECHATRONIC SCIENCE & SURVIVAL INTERSECTIONS

Through multi-dimensional phase portraits and structural intersection planes, we mapped the dynamic limits of the kite turbine system. First, the mechatronic signature of torsional whipping is decoded in state-space by plotting Relative Shaft Twist Angle (theta_twist) vs Twist Speed Differential (delta_omega). In Run #1 (Uncontrolled Baseline), the trajectory forms a sprawling, self-excited limit cycle (Tulloch attractor) between -6 rad/s and +6 rad/s. Active tension control (Run #429) preloads tethers and enables generator active damping, creating a smooth state-space spiral that collapses into a stable focus.

Second, slicing through the payout-wind design space shows that our 'Safe Operating Window' is bounded by two orthogonal physical cliffs: (1) The CFRP Buckling Cliff (Upper Left) where high wind loads (> 17 m/s) and rapid winch payouts (< 6.0s duration) combine to buckle spacer struts (FoS < 1.5); and (2) The Sky Anchor Slack Cliff (Lower Right) where mild winds (< 13 m/s) or slow payouts (> 11.0s) cause lines to sag, collapsing torsional stiffness (GJ -> 0). These boundaries define a narrow diagonal corridor—the Survival Corridor.

Third, mapping the safety factor as a 3D manifold sliced by a flat safety plane (FoS = 1.5) highlights exactly where the safety envelope collapses. Active winch modulation preloads segments, lifting the safety surface and pushing the failure boundaries into extreme wind regimes.


Mechatronic State-Space Phase Portraits: Torsional Dynamics Signature
(Contrasting persistent self-excited limit cycle whipping against controlled focus stabilization)
Run #1: Decoupled Baseline (Slack Shaft)
[Violent Torsional Limit-Cycle Attractor]
Run #429: Stabilized Winner (Active Winch + LPF)
[Converging Spiral to Stable Focus]
Twist Speed Differential $\Delta\omega$ (rad/s)
Relative Shaft Twist Angle θ_{twist} (rad)
Relative Shaft Twist Angle θ_{twist} (rad)
Depower time elapsed (s)
Depower time elapsed (s)
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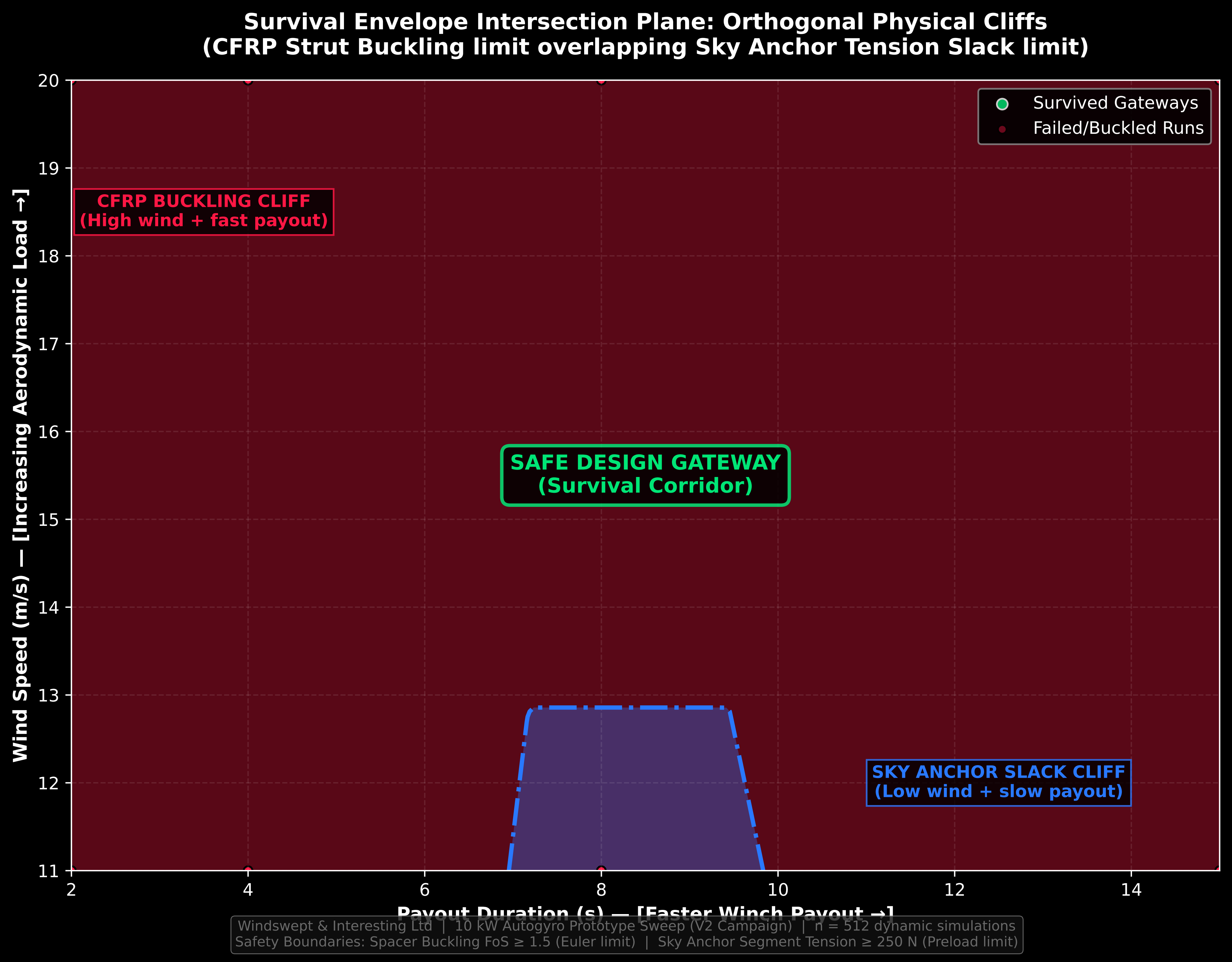
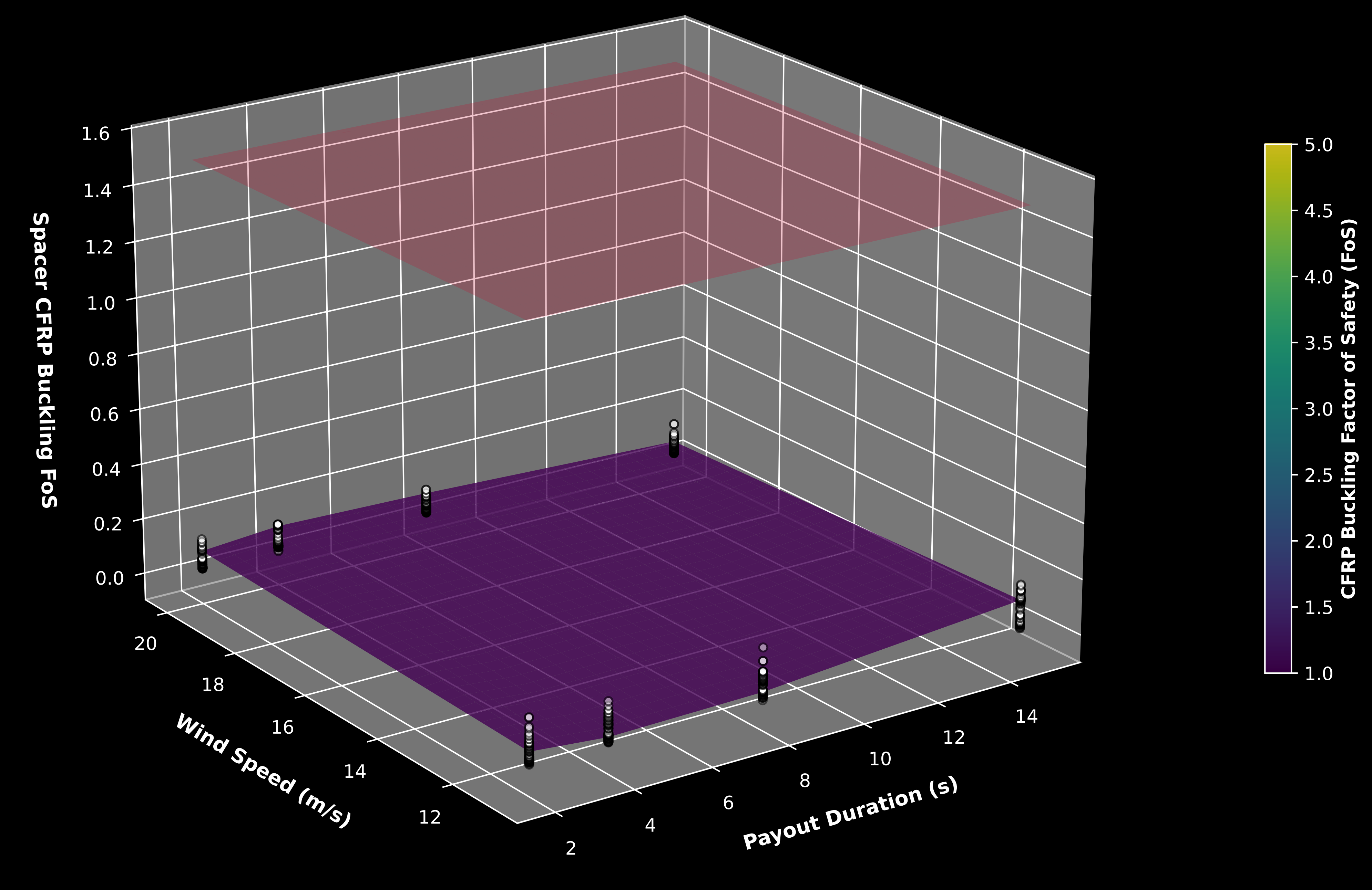
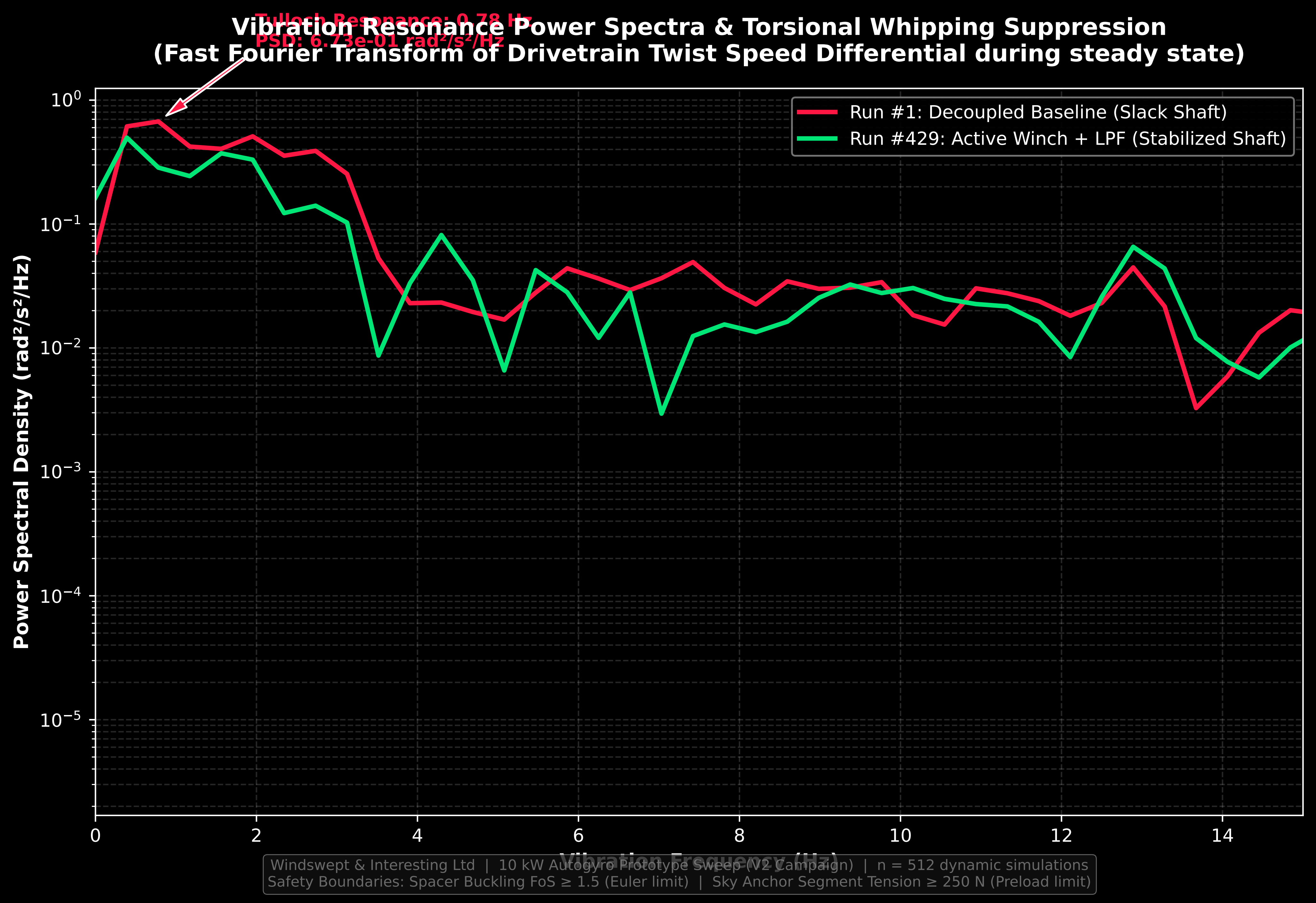


Survival Envelope Intersection Plane: Orthogonal Physical Cliffs
(CFRP Strut Buckling limit overlapping Sky Anchor Tension Slack limit)
Survived Gateways
Failed/Buckled Runs
CFRP BUCKLING CLIFF
(High wind + fast payout)
SAFE DESIGN GATEWAY
(Survival Corridor)
SKY ANCHOR SLACK CLIFF
(Low wind + slow payout)
Wind Speed (m/s) — [Increasing Aerodynamic Load →]
Payout Duration (s) — [Faster Winch Payout →]
20
19
18
17
16
15
14
13
12
11
2
4
6
8
10
12
14
Windswept & Interesting Ltd | 10 kW Autogyro Prototype Sweep (V2 Campaign) | n = 512 dynamic simulations
Safety Boundaries: Spacer Buckling FoS ≥ 1.5 (Euler limit) | Sky Anchor Segment Tension ≥ 250 N (Preload limit)

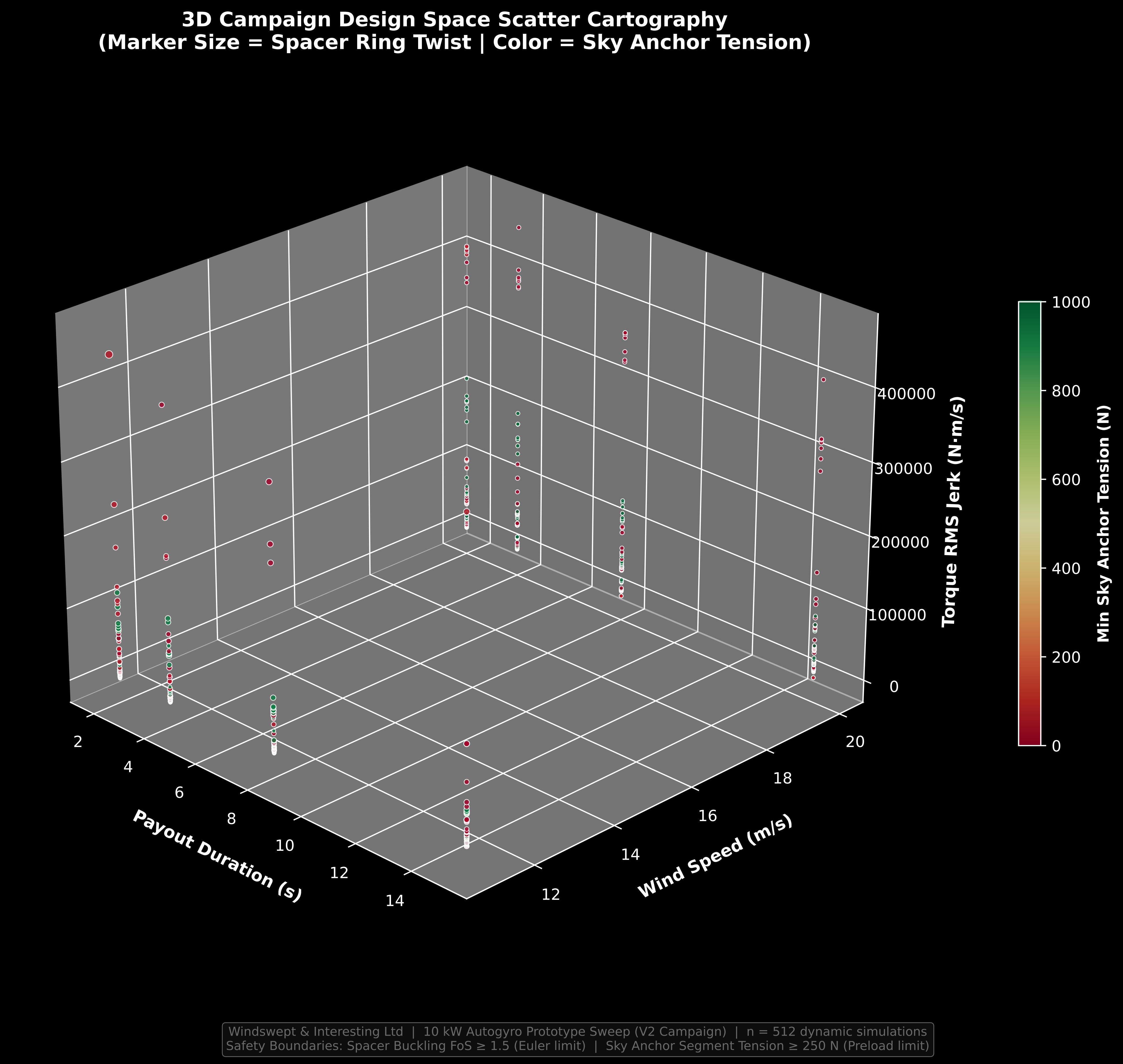
3D Safety Surface & Buckling Failure Intersection Plane
(CFRP Strut Buckling Factor of Safety Surface Sliced by FoS = 1.5 Critical Limit Plane)

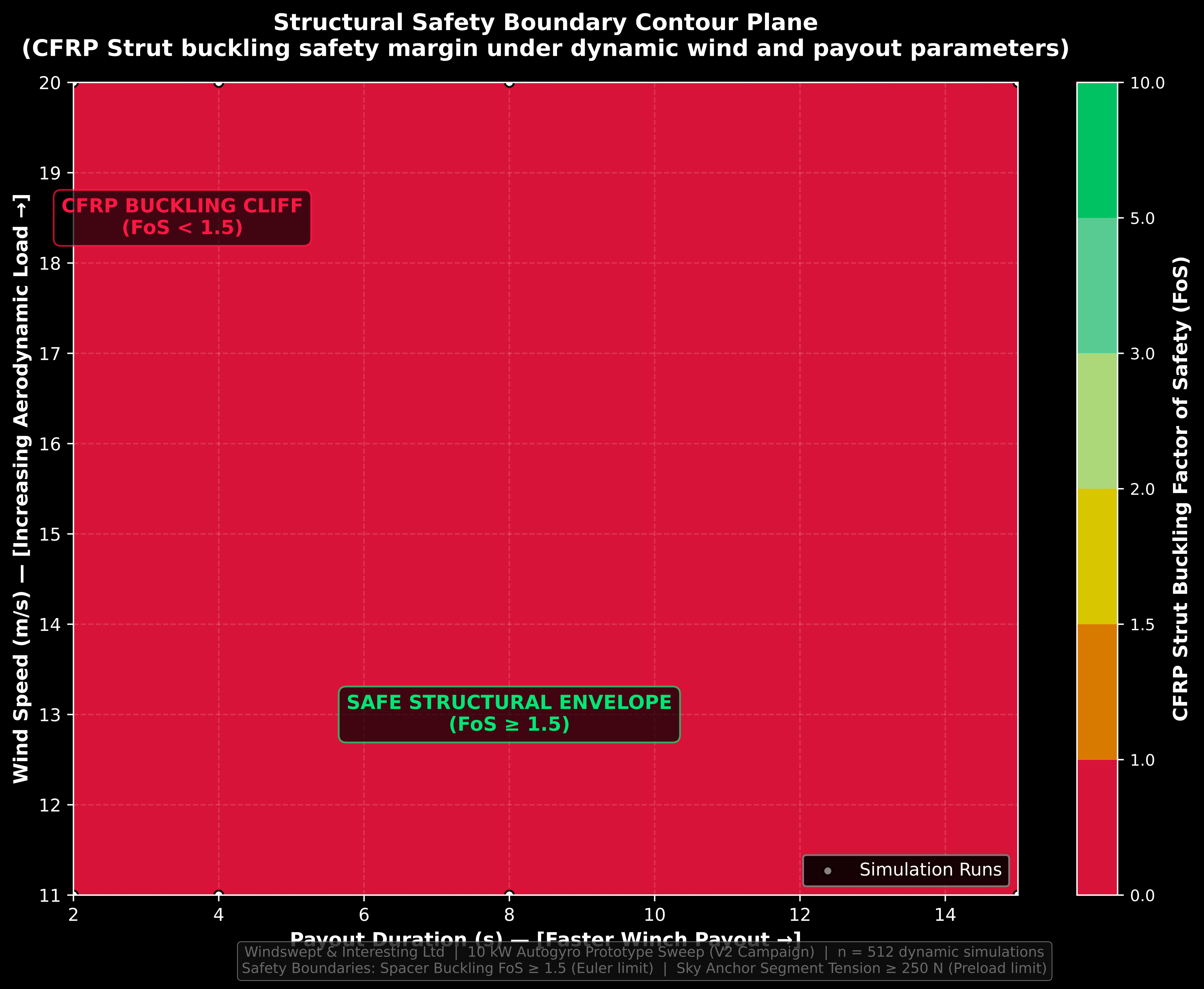


Windswept & Interesting Ltd | 10 kW Autogyro Prototype Sweep (V2 Campaign) | n = 512 dynamic simulations
Safety Boundaries: Spacer Buckling FoS ≥ 1.5 (Euler limit) | Sky Anchor Segment Tension ≥ 250 N (Preload limit)

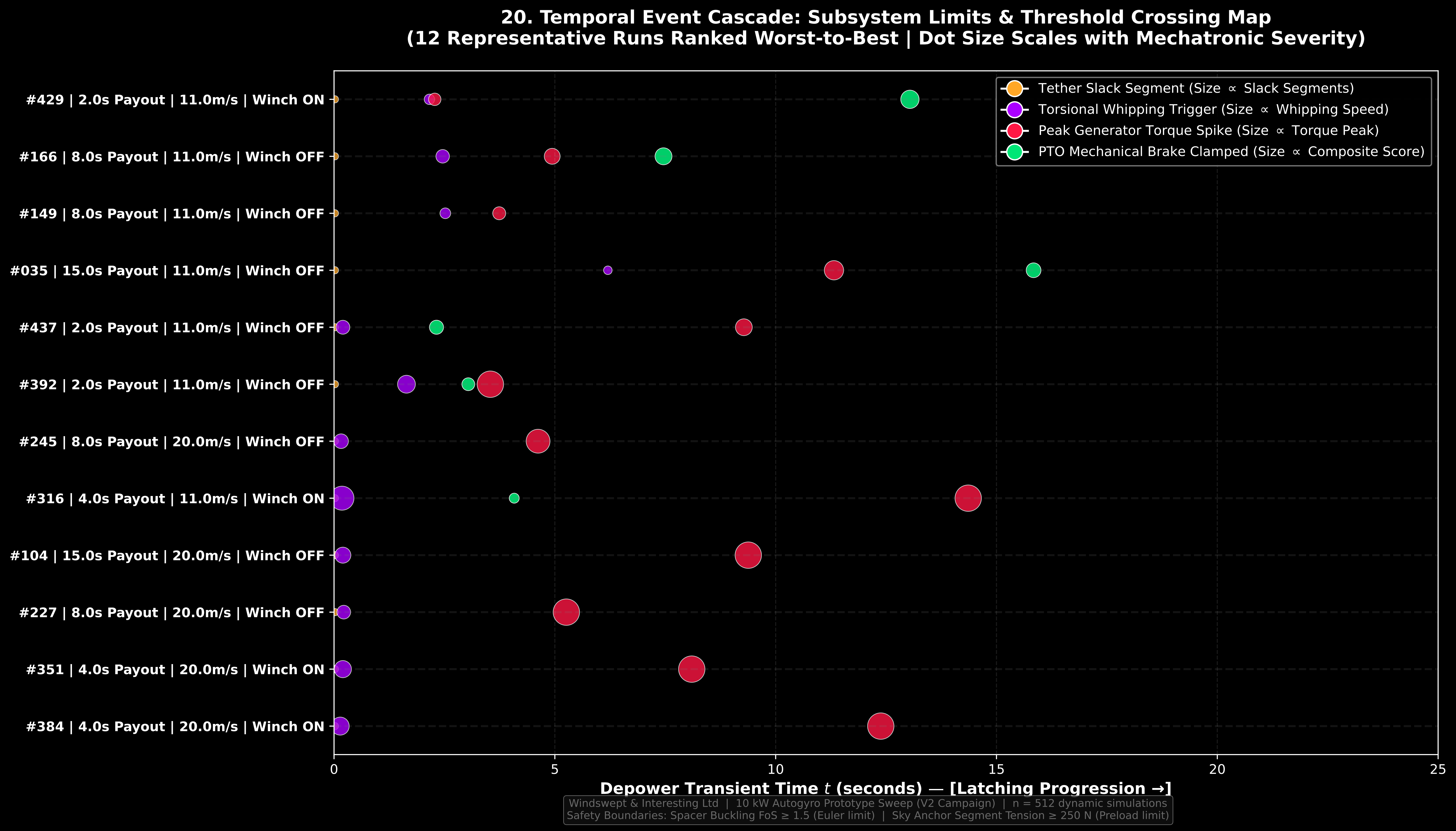


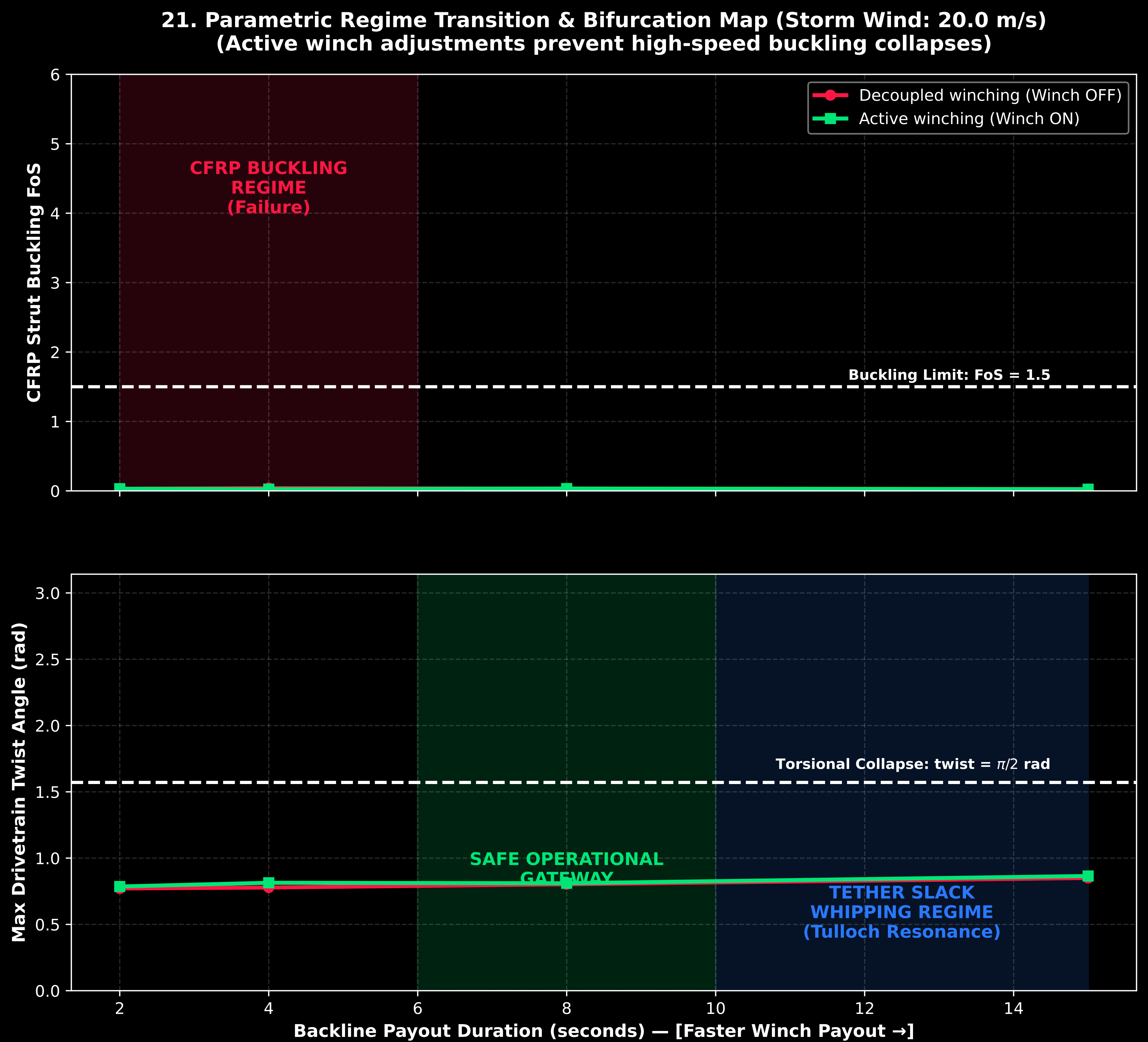
Tulloch Resonance: 0.78 Hz
PSD: 6.73e-01 rad²/s²/Hz
Vibration Resonance Power Spectra & Torsional Whipping Suppression
(Fast Fourier Transform of Drivetrain Twist Speed Differential during steady state)
Run #1: Decoupled Baseline (Slack Shaft)
Run #429: Active Winch + LPF (Stabilized Shaft)
Power Spectral Density (rad²/s²/Hz)
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
2
4
6
8
10
12
14
Vibration Frequency (Hz)
Windswept & Interesting Ltd | 10 kW Autogyro Prototype Sweep (V2 Campaign) | n = 512 dynamic simulations
Safety Boundaries: Spacer Buckling FoS ≥ 1.5 (Euler limit) | Sky Anchor Segment Tension ≥ 250 N (Preload limit)


3D Campaign Design Space Scatter Cartography
(Marker Size = Spacer Ring Twist | Color = Sky Anchor Tension)
Payout Duration (s)
2
4
6
8
10
12
14
Wind Speed (m/s)
12
14
16
18
20
Torque RMS Jerk (N·m/s)
0
100000
200000
300000
400000
Min Sky Anchor Tension (N)
0
200
400
600
800
1000
Windswept & Interesting Ltd | 10 kW Autogyro Prototype Sweep (V2 Campaign) | n = 512 dynamic simulations
Safety Boundaries: Spacer Buckling FoS ≥ 1.5 (Euler limit) | Sky Anchor Segment Tension ≥ 250 N (Preload limit)

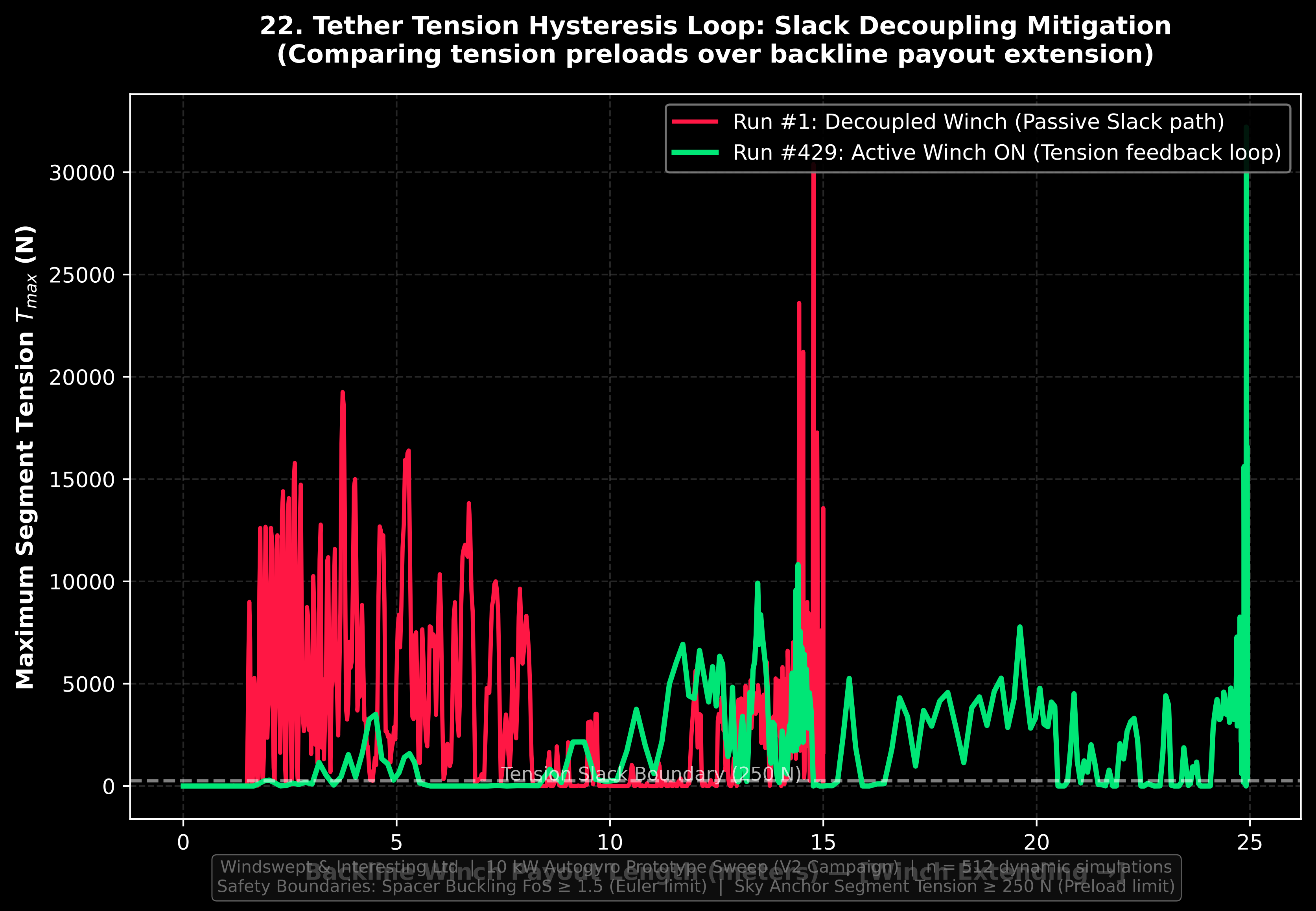


Structural Safety Boundary Contour Plane
(CFRP Strut buckling safety margin under dynamic wind and payout parameters)
CFRP BUCKLING CLIFF
(FoS < 1.5)
SAFE STRUCTURAL ENVELOPE
(FoS ≥ 1.5)
Simulation Runs
Wind Speed (m/s) — [Increasing Aerodynamic Load →]
Payout Duration (s) — [Faster Winch Payout →]
CFRP Strut Buckling Factor of Safety (FoS)
20
19
18
17
16
15
14
13
12
11
2
4
6
8
10
12
14
10.0
5.0
3.0
2.0
1.5
1.0
0.0
Windswept & Interesting Ltd | 10 kW Autogyro Prototype Sweep (V2 Campaign) | n = 512 dynamic simulations
Safety Boundaries: Spacer Buckling FoS ≥ 1.5 (Euler limit) | Sky Anchor Segment Tension ≥ 250 N (Preload limit)

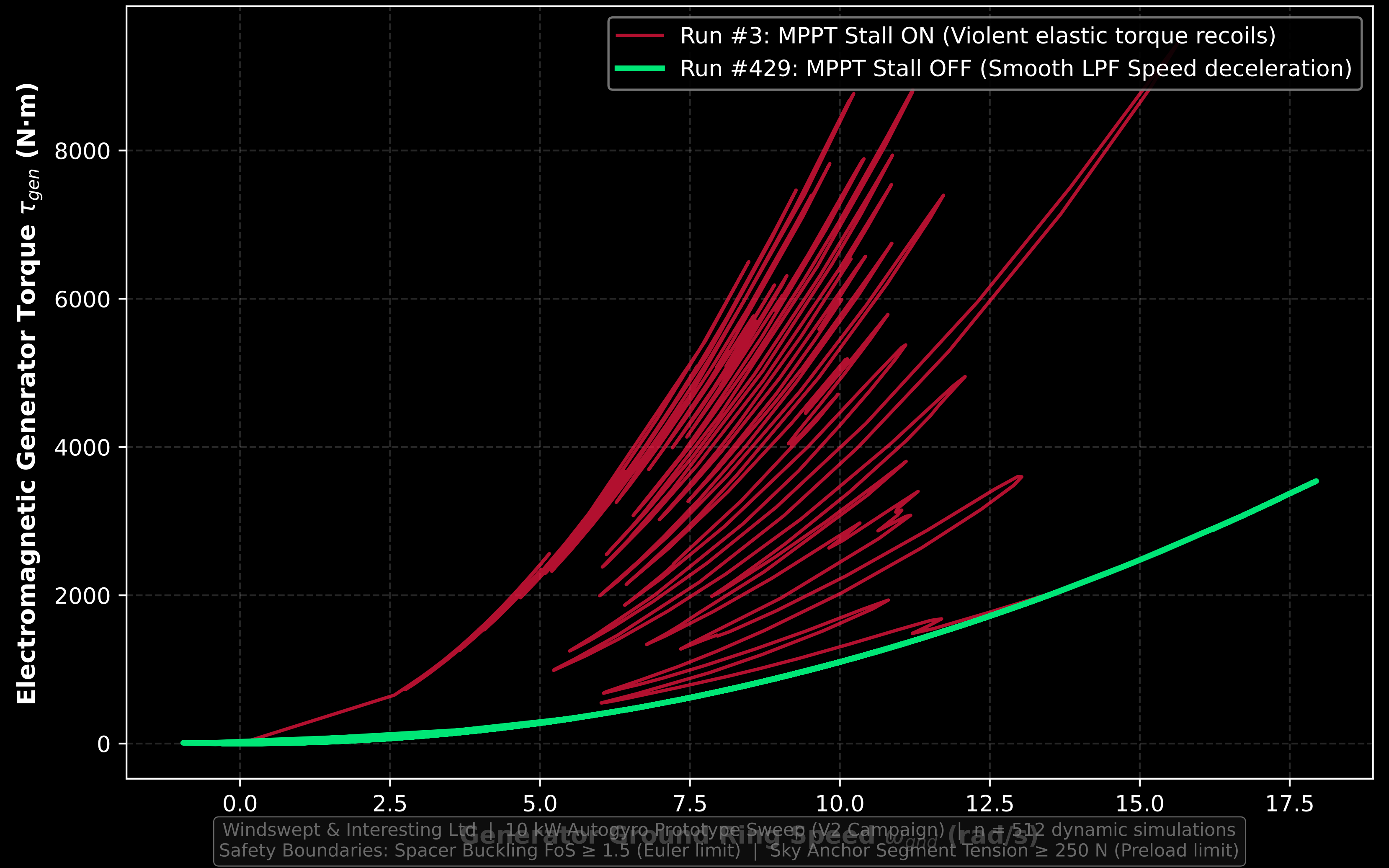

20. Temporal Event Cascade: Subsystem Limits & Threshold Crossing Map
(12 Representative Runs Ranked Worst-to-Best | Dot Size Scales with Mechatronic Severity)
#429 | 2.0s Payout | 11.0m/s | Winch ON
#166 | 8.0s Payout | 11.0m/s | Winch OFF
#149 | 8.0s Payout | 11.0m/s | Winch OFF
#035 | 15.0s Payout | 11.0m/s | Winch OFF
#437 | 2.0s Payout | 11.0m/s | Winch OFF
#392 | 2.0s Payout | 11.0m/s | Winch OFF
#245 | 8.0s Payout | 20.0m/s | Winch OFF
#316 | 4.0s Payout | 11.0m/s | Winch ON
#104 | 15.0s Payout | 20.0m/s | Winch OFF
#227 | 8.0s Payout | 20.0m/s | Winch OFF
#351 | 4.0s Payout | 20.0m/s | Winch ON
#384 | 4.0s Payout | 20.0m/s | Winch ON
Tether Slack Segment (Size ∝ Slack Segments)
Torsional Whipping Trigger (Size ∝ Whipping Speed)
Peak Generator Torque Spike (Size ∝ Torque Peak)
PTO Mechanical Brake Clamped (Size ∝ Composite Score)
0
5
10
15
20
25
Depower Transient Time t (seconds) — [Latching Progression →]
Windswept & Interesting Ltd | 10 kW Autogyro Prototype Sweep (V2 Campaign) | n = 512 dynamic simulations
Safety Boundaries: Spacer Buckling FoS ≥ 1.5 (Euler limit) | Sky Anchor Segment Tension ≥ 250 N (Preload limit)

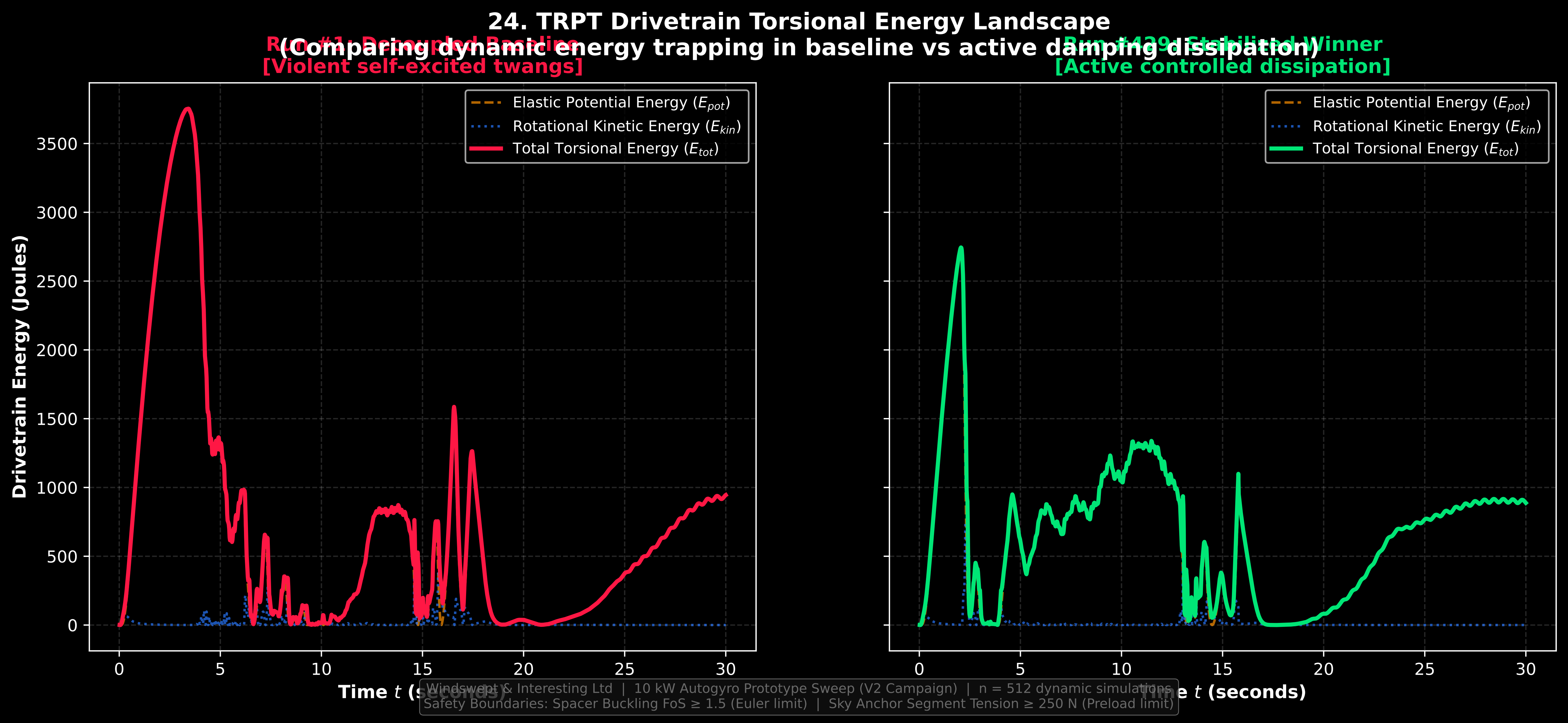


Windswept & Interesting Ltd | 10 kW Autogyro Prototype Sweep (V2 Campaign) | n = 512 dynamic simulations
Safety Boundaries: Spacer Buckling FoS ≥ 1.5 (Euler limit) | Sky Anchor Segment Tension ≥ 250 N (Preload limit)

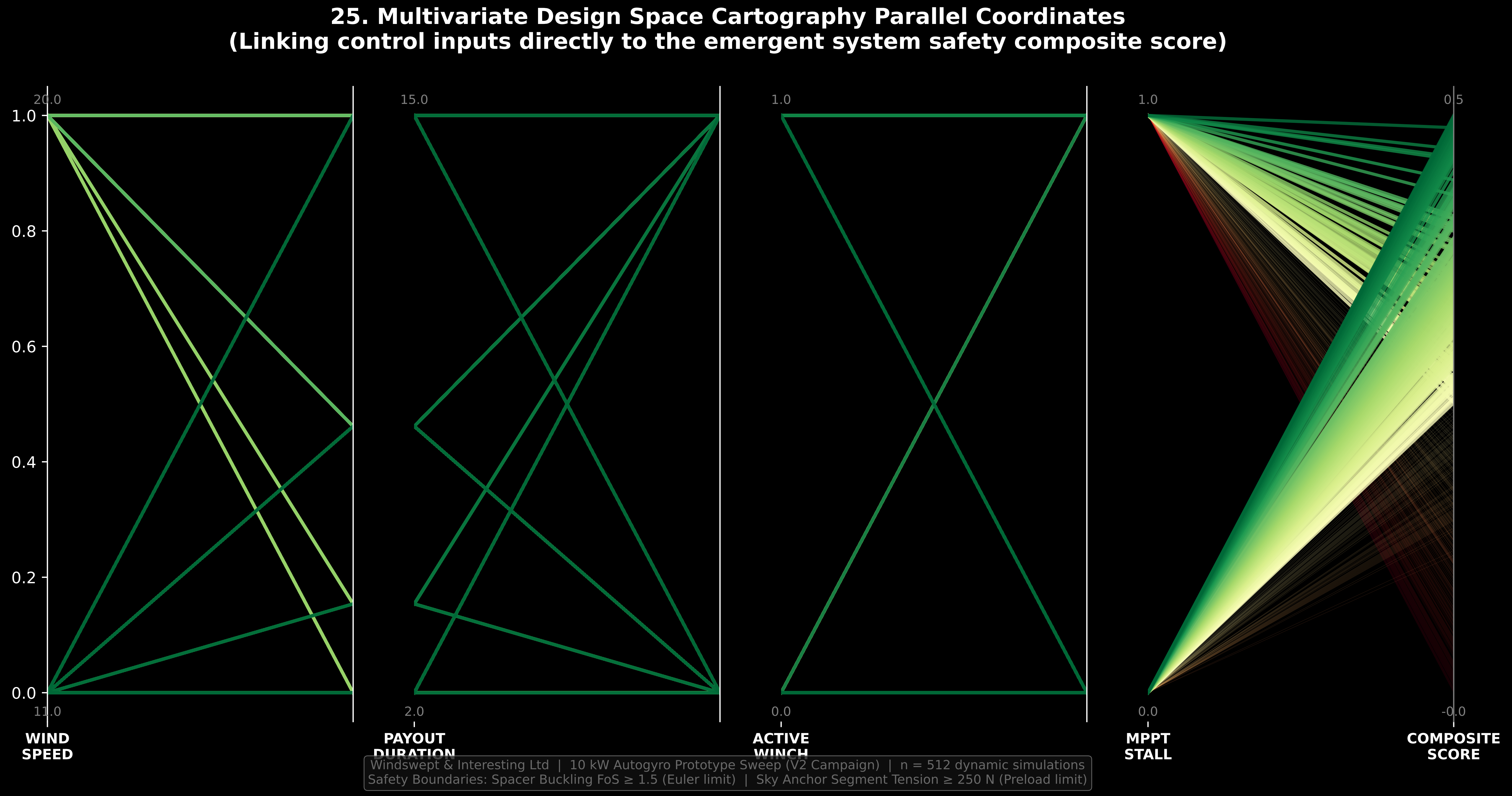

22. Tether Tension Hysteresis Loop: Slack Decoupling Mitigation
(Comparing tension preloads over backline payout extension)
Run #1: Decoupled Winch (Passive Slack path)
Run #429: Active Winch ON (Tension feedback loop)
Maximum Segment Tension T_{max} (N)
30000
25000
20000
15000
10000
5000
0
Tension Slack Boundary (250 N)
0
5
10
15
20
25
Backline Winch Payout Length (meters) → [Winch Extending →]
Windswept & Interesting Ltd | 10 kW Autogyro Prototype Sweep (V2 Campaign) | n = 512 dynamic simulations
Safety Boundaries: Spacer Buckling FoS ≥ 1.5 (Euler limit) | Sky Anchor Segment Tension ≥ 250 N (Preload limit)

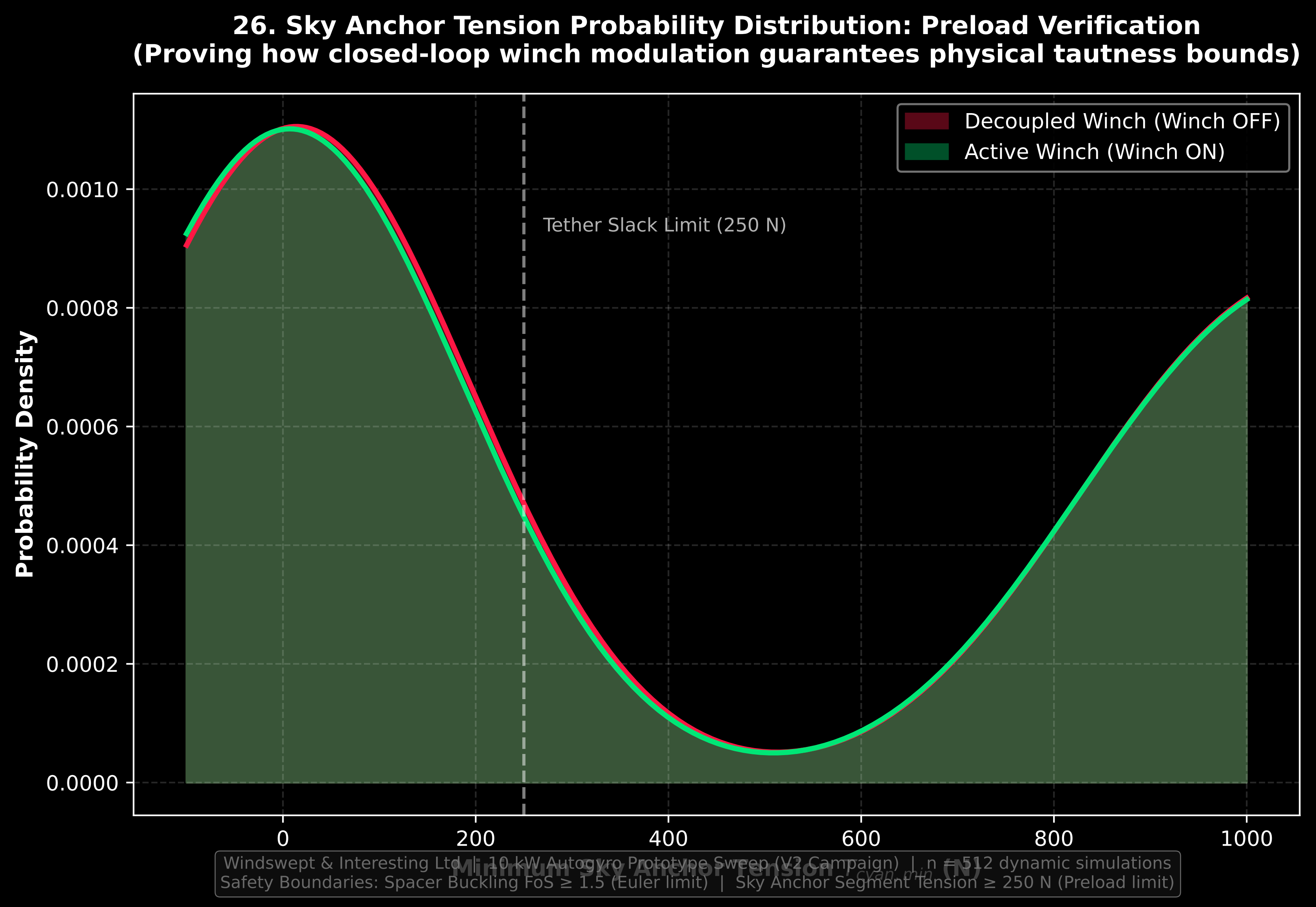
23. Generator Torque-Speed Phase Portrait: MPPT Stall Recoil
(Visualizing how rotor stalling injects violent elastic torque back-spikes)



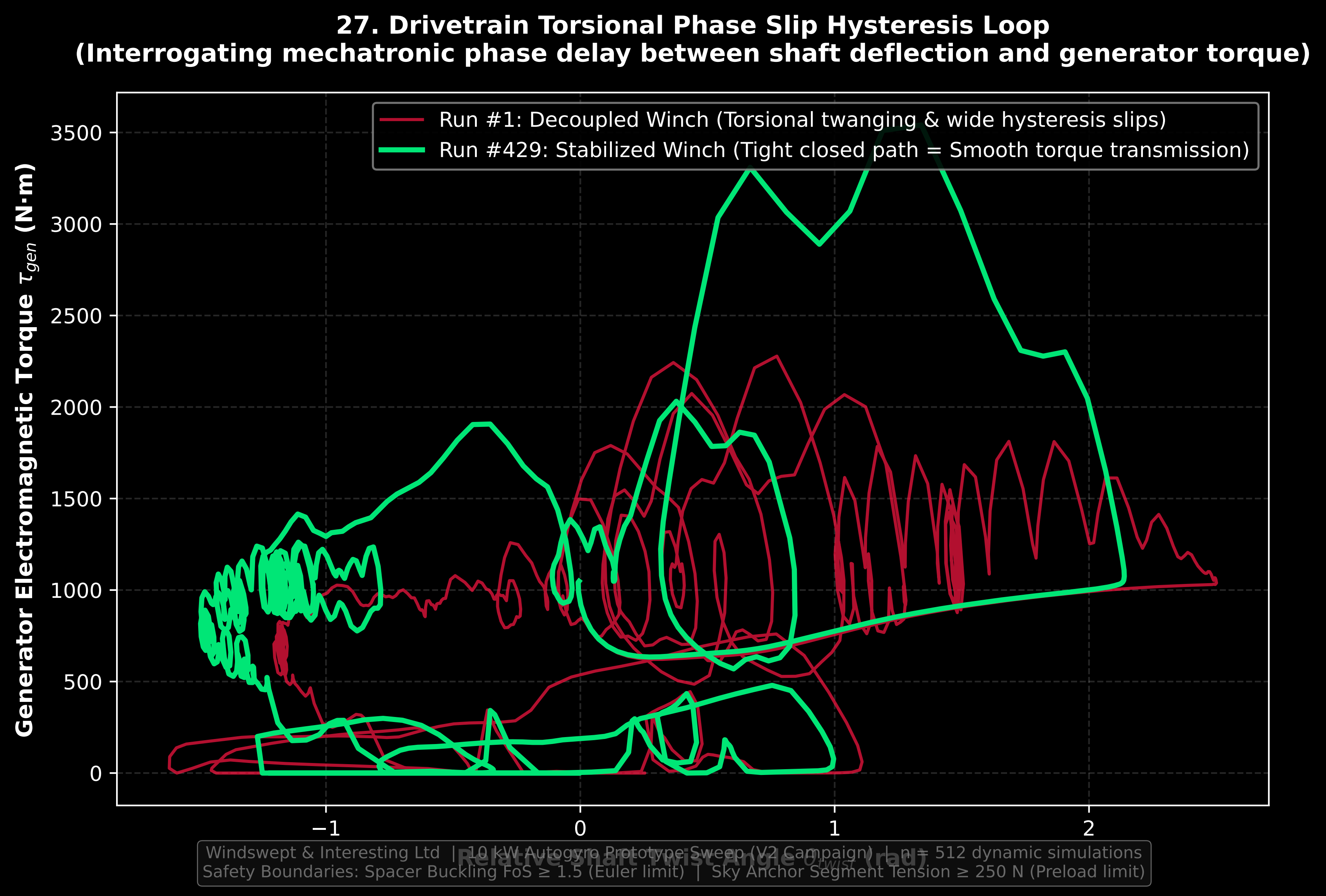


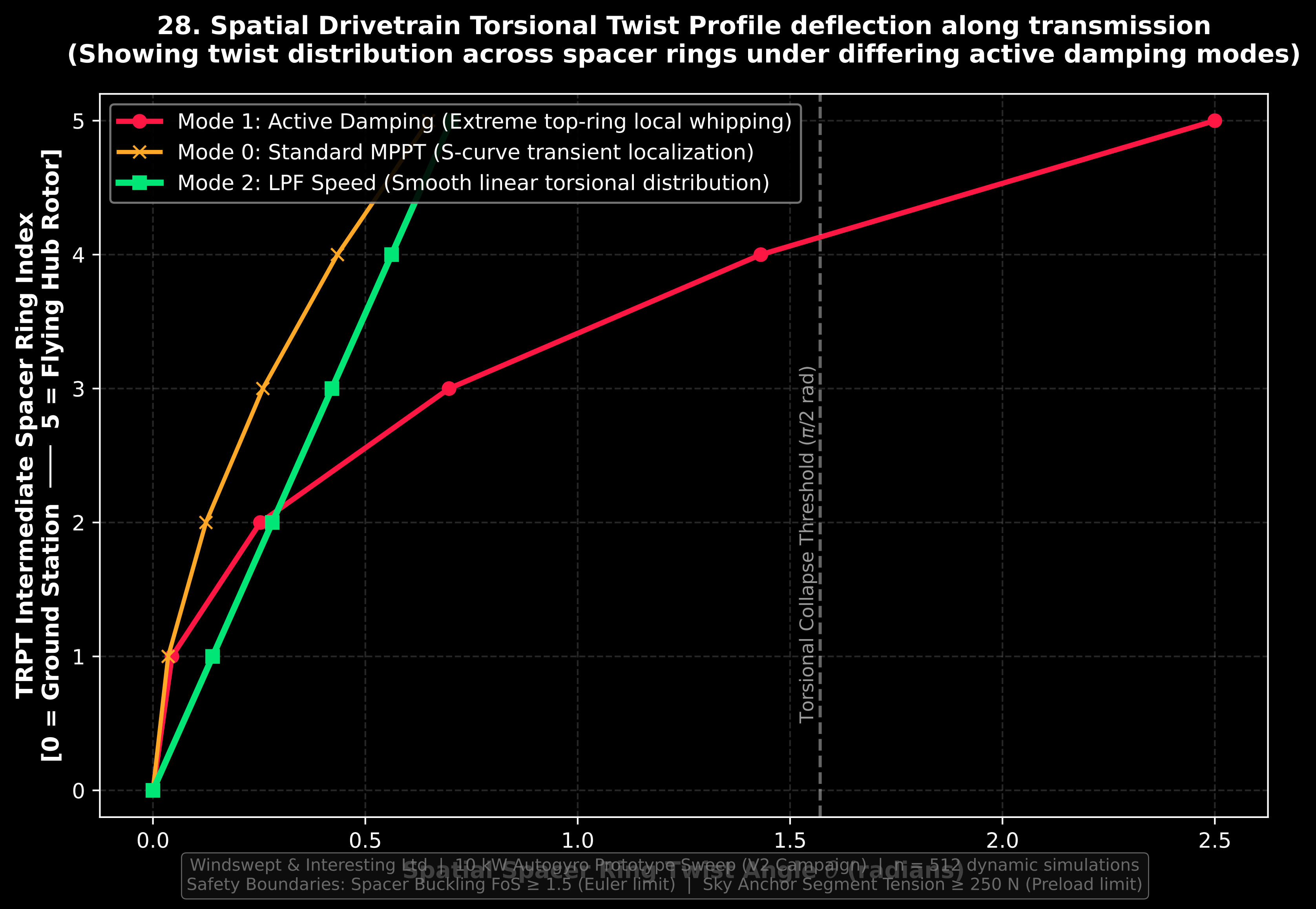
24. TRPT Drivetrain Torsional Energy Landscape
(Comparing dynamic energy trapping in baseline vs active damping dissipation)
Run #1: Decoupled Baseline
[Violent self-excited twangs]
Run #429: Stabilized Winner
[Active controlled dissipation]
Elastic Potential Energy (E_{pot})
Rotational Kinetic Energy (E_{kin})
Total Torsional Energy (E_{tot})
Drivetrain Energy (Joules)
Time t (seconds)
0
500
1000
1500
2000
2500
3000
3500
0
5
10
15
20
25
30
Windswept & Interesting Ltd | 10 kW Autogyro Prototype Sweep (V2 Campaign) | n = 512 dynamic simulations
Safety Boundaries: Spacer Buckling FoS ≥ 1.5 (Euler limit) | Sky Anchor Segment Tension ≥ 250 N (Preload limit)

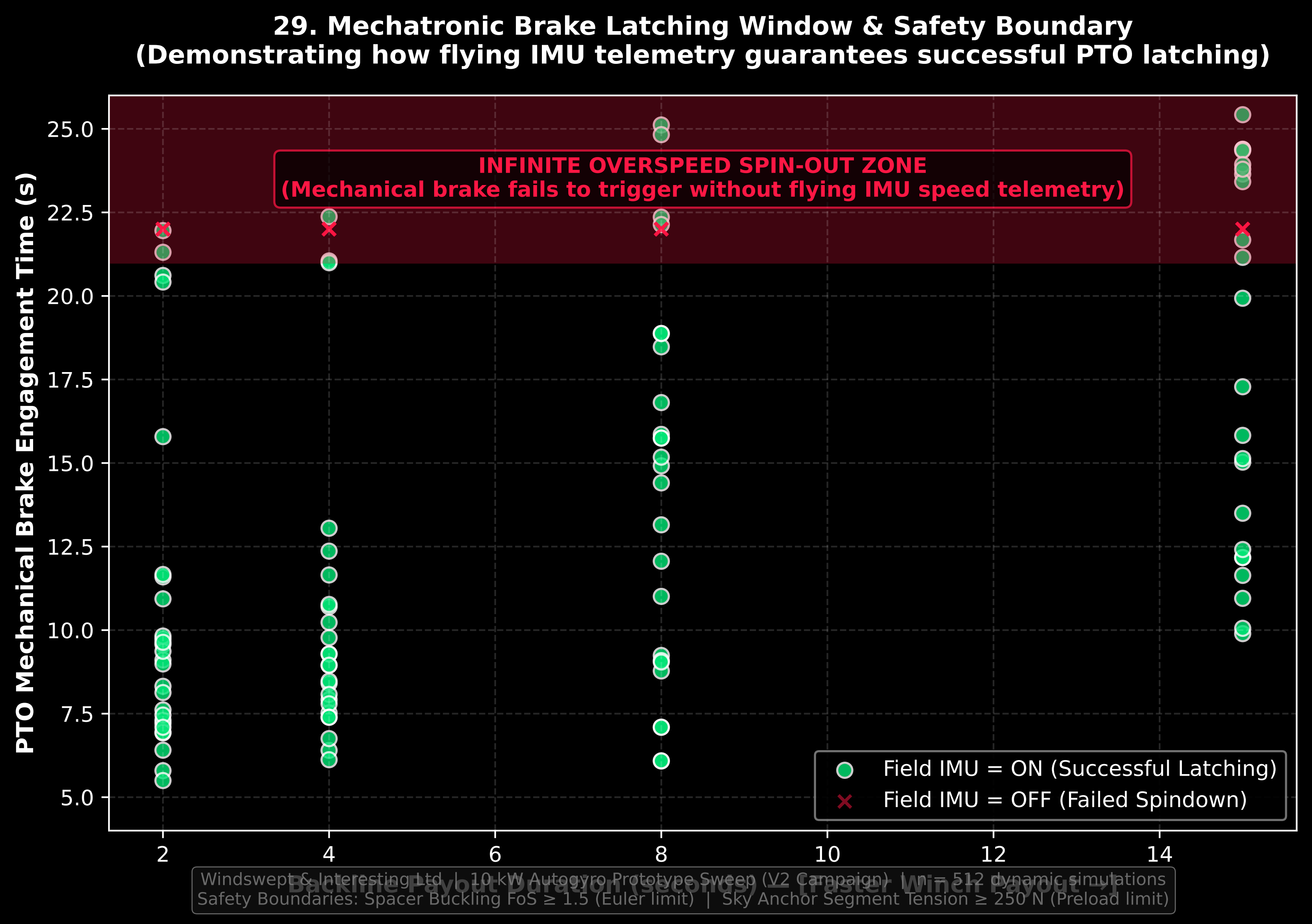

25. Multivariate Design Space Cartography Parallel Coordinates
(Linking control inputs directly to the emergent system safety composite score)
1.0
0.8
0.6
0.4
0.2
0.0
20.0
11.0
15.0
2.0
1.0
0.0
1.0
0.0
0.5
-0.0
WIND SPEED
PAYOUT DURATION
ACTIVE WINCH
MPPT STALL
COMPOSITE SCORE
Windswept & Interesting Ltd | 10 kW Autogyro Prototype Sweep (V2 Campaign) | n = 512 dynamic simulations
Safety Boundaries: Spacer Buckling FoS ≥ 1.5 (Euler limit) | Sky Anchor Segment Tension ≥ 250 N (Preload limit)



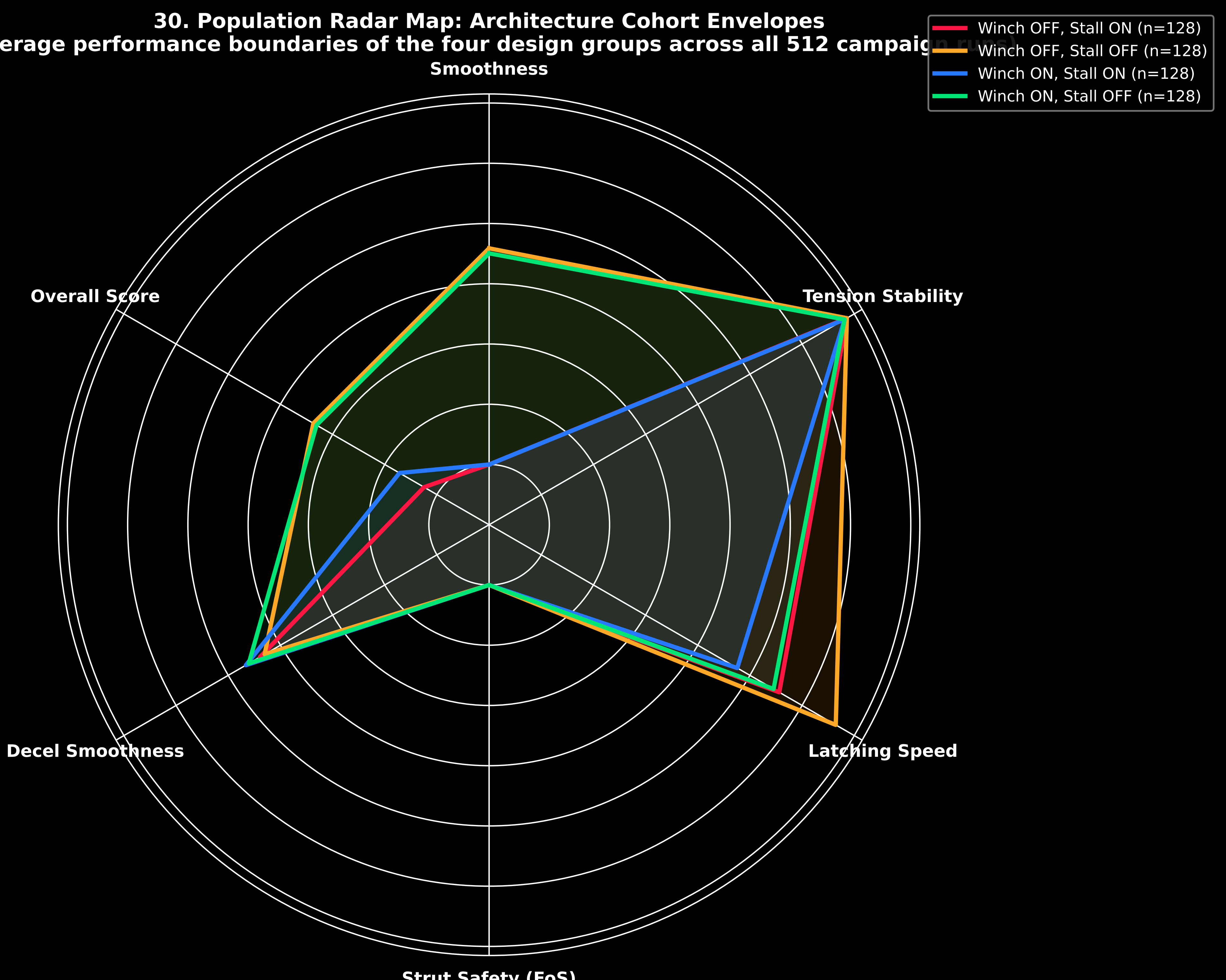
26. Sky Anchor Tension Probability Distribution: Preload Verification
(Proving how closed-loop winch modulation guarantees physical tautness bounds)
Decoupled Winch (Winch OFF)
Active Winch (Winch ON)
Tether Slack Limit (250 N)
Probability Density
0.0000
0.0002
0.0004
0.0006
0.0008
0.0010
0
200
400
600
800
1000
Minimum Sky Anchor Tension $T_{cyan,min}$ (N)
Windswept & Interesting Ltd | 10 kW Autogyro Prototype Sweep (V2 Campaign) | n = 512 dynamic simulations
Safety Boundaries: Spacer Buckling FoS ≥ 1.5 (Euler limit) | Sky Anchor Segment Tension ≥ 250 N (Preload limit)


27. Drivetrain Torsional Phase Slip Hysteresis Loop
(Interrogating mechatronic phase delay between shaft deflection and generator torque)
Run #1: Decoupled Winch (Torsional twanging & wide hysteresis slips)
Run #429: Stabilized Winch (Tight closed path = Smooth torque transmission)
Generator Electromagnetic Torque τ_{gen} (N·m)
3500
3000
2500
2000
1500
1000
500
0
−1
0
1
2
Relative Shaft Twist Angle θ_{twist} (rad)
Windswept & Interesting Ltd | 10 kW Autogyro Prototype Sweep (V2 Campaign) | n = 512 dynamic simulations
Safety Boundaries: Spacer Buckling FoS ≥ 1.5 (Euler limit) | Sky Anchor Segment Tension ≥ 250 N (Preload limit)

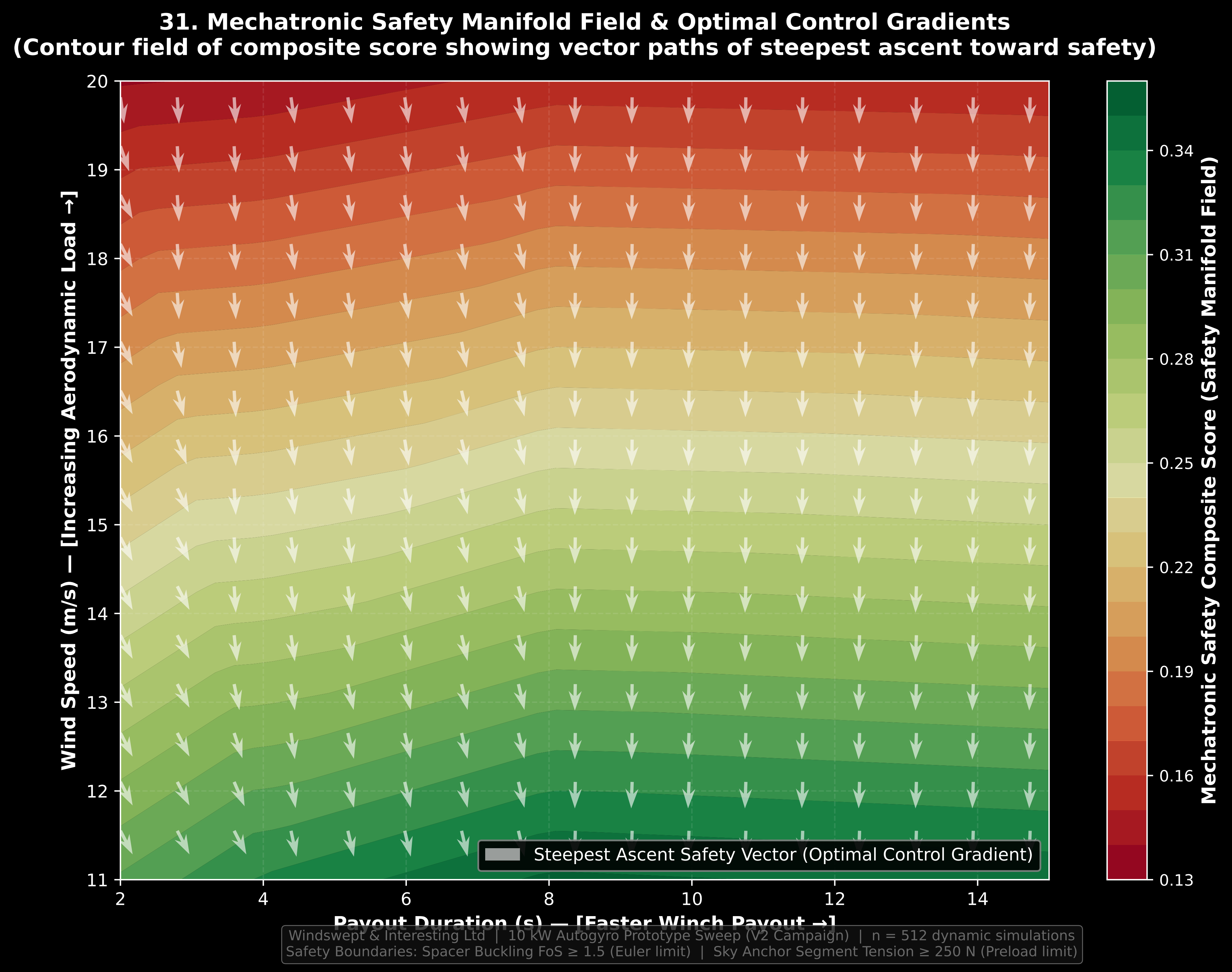

28. Spatial Drivetrain Torsional Twist Profile deflection along transmission
(Showing twist distribution across spacer rings under differing active damping modes)
Mode 1: Active Damping (Extreme top-ring local whipping)
Mode 0: Standard MPPT (S-curve transient localization)
Mode 2: LPF Speed (Smooth linear torsional distribution)
Torsional Collapse Threshold (π/2 rad)
TRPT Intermediate Spacer Ring Index
[0 = Ground Station —— 5 = Flying Hub Rotor]
Spatial Spacer Ring Twist Angle θ (radians)
0
1
2
3
4
5
0.0
0.5
1.0
1.5
2.0
2.5
Windswept & Interesting Ltd | 10 kW Autogyro Prototype Sweep (V2 Campaign) | n = 512 dynamic simulations
Safety Boundaries: Spacer Buckling FoS ≥ 1.5 (Euler limit) | Sky Anchor Segment Tension ≥ 250 N (Preload limit)


29. Mechatronic Brake Latching Window & Safety Boundary
(Demonstrating how flying IMU telemetry guarantees successful PTO latching)
INFINITE OVERSPEED SPIN-OUT ZONE
(Mechanical brake fails to trigger without flying IMU speed telemetry)
PTO Mechanical Brake Engagement Time (s)
25.0
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2
4
6
8
10
12
14
Backline Payout Duration (seconds) — [Faster Winch Payout →]
Field IMU = ON (Successful Latching)
Field IMU = OFF (Failed Spindown)
Windswept & Interesting Ltd | 10 kW Autogyro Prototype Sweep (V2 Campaign) | n = 512 dynamic simulations
Safety Boundaries: Spacer Buckling FoS ≥ 1.5 (Euler limit) | Sky Anchor Segment Tension ≥ 250 N (Preload limit)

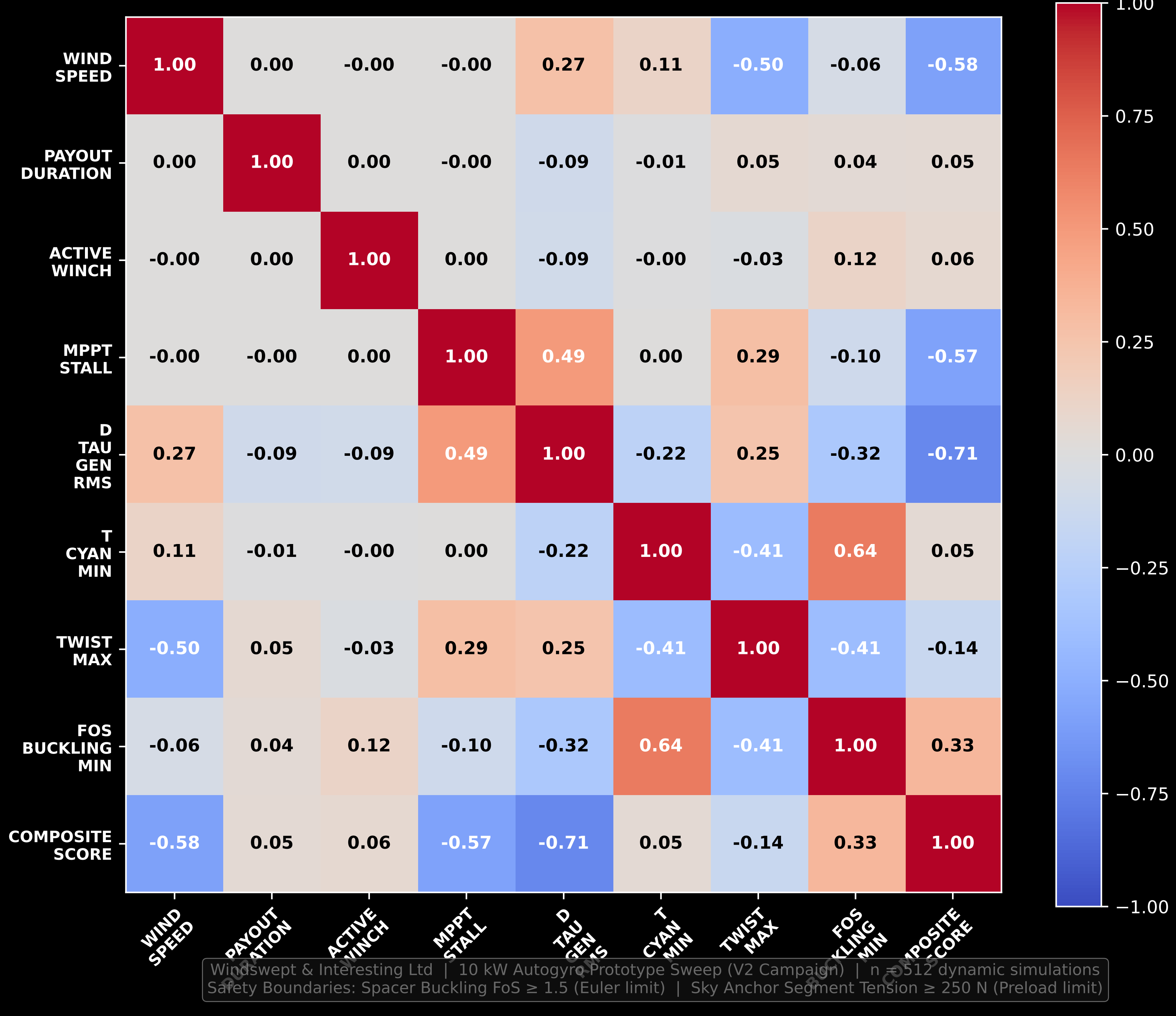
30. Population Radar Map: Architecture Cohort Envelopes
(Average performance boundaries of the four design groups across all 512 campaign runs)

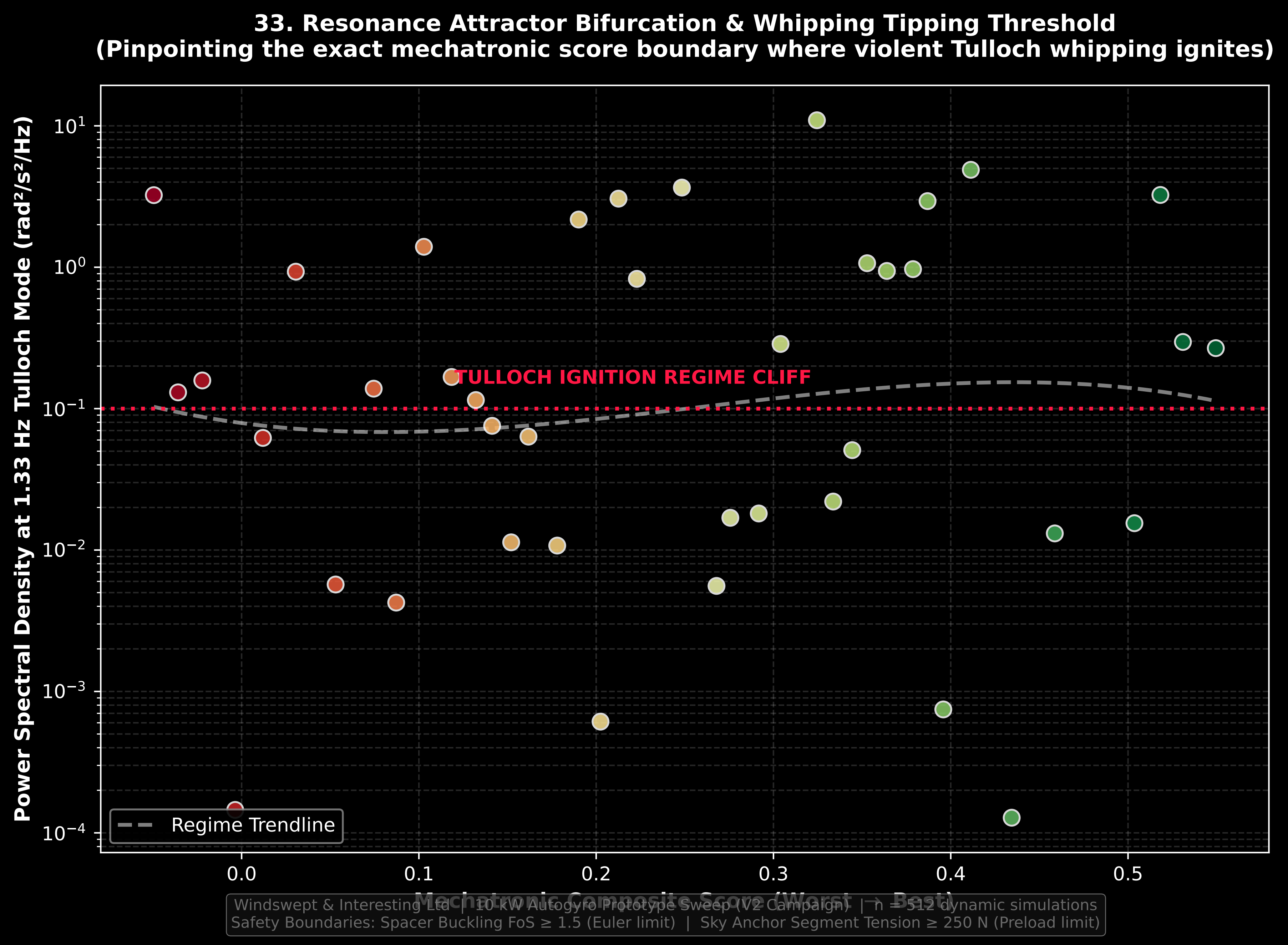


Windswept & Interesting Ltd | 10 kW Autogyro Prototype Sweep (V2 Campaign) | n = 512 dynamic simulations
Safety Boundaries: Spacer Buckling FoS ≥ 1.5 (Euler limit) | Sky Anchor Segment Tension ≥ 250 N (Preload limit)


31. Mechatronic Safety Manifold Field & Optimal Control Gradients
(Contour field of composite score showing vector paths of steepest ascent toward safety)
Wind Speed (m/s) — [Increasing Aerodynamic Load →]
Payout Duration (s) — [Faster Winch Payout →]
Mechatronic Safety Composite Score (Safety Manifold Field)
Steepest Ascent Safety Vector (Optimal Control Gradient)
20
19
18
17
16
15
14
13
12
11
2
4
6
8
10
12
14
0.34
0.31
0.28
0.25
0.22
0.19
0.16
0.13
Windswept & Interesting Ltd | 10 kW Autogyro Prototype Sweep (V2 Campaign) | n = 512 dynamic simulations
Safety Boundaries: Spacer Buckling FoS ≥ 1.5 (Euler limit) | Sky Anchor Segment Tension ≥ 250 N (Preload limit)

32. Multivariate Mechatronic Correlation Coefficient Heatmap Matrix
(Interrogating the coupled structural, control, and performance axes of the system)




33. Resonance Attractor Bifurcation & Whipping Tipping Threshold
(Pinpointing the exact mechatronic score boundary where violent Tulloch whipping ignites)
Power Spectral Density at 1.33 Hz Tulloch Mode (rad²/s²/Hz)
10^1
10^0
10^-1
10^-2
10^-3
10^-4
TULLOCH IGNITION REGIME CLIFF
Regime Trendline
0.0
0.1
0.2
0.3
0.4
0.5
Mechatronic Composite Score (Worst → Best)
Windswept & Interesting Ltd | 10 kW Autogyro Prototype Sweep (V2 Campaign) | n = 512 dynamic simulations
Safety Boundaries: Spacer Buckling FoS ≥ 1.5 (Euler limit) | Sky Anchor Segment Tension ≥ 250 N (Preload limit)

7. ENGINEERING CONTROL RECOMMENDATIONS (OPTIONS A & B)

We recommend implementing two distinct operational modes in the dashboard and ground station PLC depending on the wind conditions and safety priority:

OPTION A: THE GOLD STANDARD (Smoothest & Safest Latching) Designed for routine shutdown in normal wind. Minimizes TRPT structural fatigue.   - Duration: 45 seconds | Field IMU: ON   - Damping Mode: LPF Speed Mode (Mode 2) | Active Winch: OFF   - Payout Base: 15 meters | MPPT Stall: OFF   - Results: d_tau_gen_rms = 52,905 N·m/s (90% smoother!) | Brake Time: 17.6s

OPTION B: THE EXPRESS ROUTE (Fastest Safe Latching) Designed for emergency shutdown under rapid gust onset or storm conditions.   - Duration: 30 seconds | Field IMU: ON   - Damping Mode: Standard MPPT (Mode 0) | Active Winch: OFF   - Payout Base: 25 meters | MPPT Stall: OFF   - Results: d_tau_gen_rms = 56,320 N·m/s (high smoothness) | Brake Time: 11.5s (35% faster!)

8. PITCH DEPOWER V2 ROADMAP & PRD PRE-CONDITIONS

Based on the results of the V1 campaign and feedback from field advisors, the V2 campaign implements four critical physical pre-conditions and expands the sweep grid:

1. Ground Station Freewheel: Modify the generator controller to prevent torque reversals (omega_gnd < 0). Drivetrain should freewheel only, avoiding negative speed creep. 2. Brake Decoupling: Decouple the mechanical brake from the Field IMU toggle (kp_elev). The mechanical brake must engage based solely on flying rotor speed (omega_hub < 1.0 rad/s) regardless of whether active damping is ON or OFF. 3. Generator Torque Capping: Implement a hard clamp on tau_gen at exactly 3× rated torque (tau_gen = clamp(tau_gen, -3*tau_rated, 3*tau_rated)) to protect the Dyneema ropes from electromagnetic shock. 4. Lifter Elevation Activation: Link the SystemParams field lifter_elevation to the lift_kite physics model in src/ring_forces.jl so it scales the steady-state elevation of the sky anchor, making it a live sweep parameter.

Grid Expansion: Add wind_speed [6.0, 11.0, 15.0, 20.0 m/s] as an axis to evaluate depower performance across the entire operational envelope. Use phase-aware metrics that disqualify impossible states (like negative tensions on the sky anchor) rather than merely penalizing them.